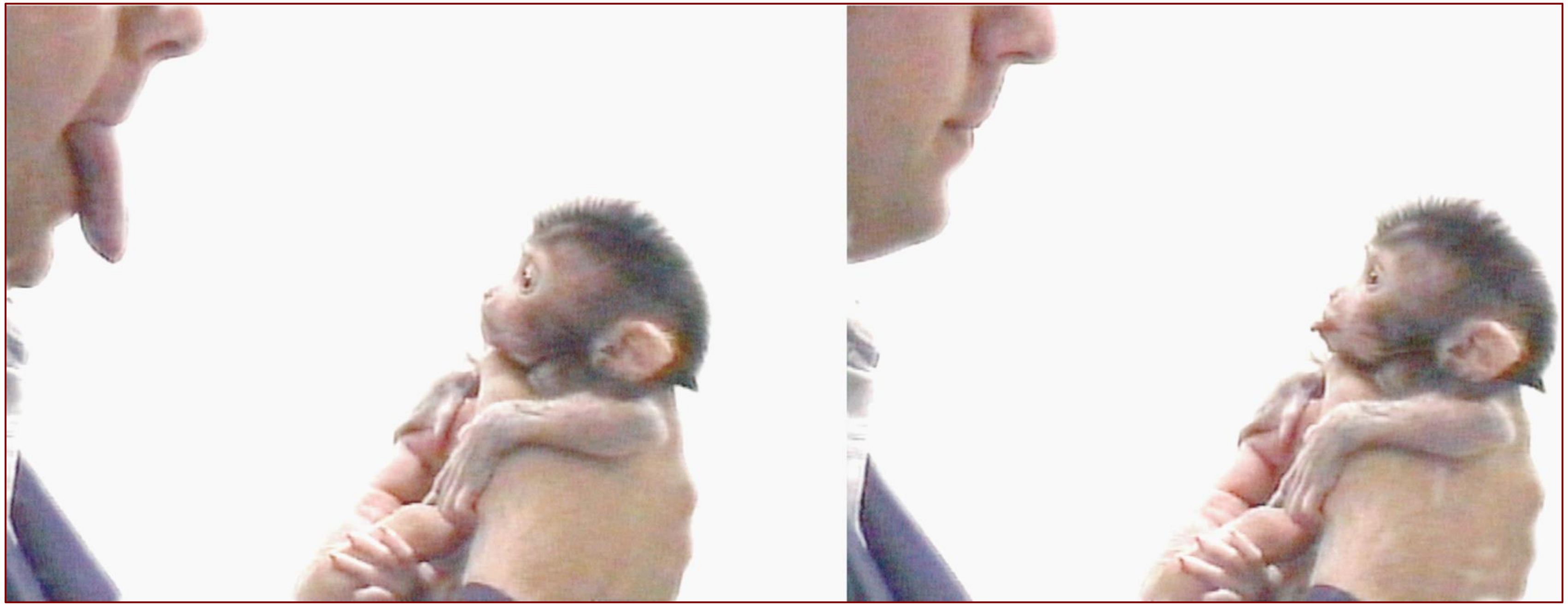
16-831: Introduction to Robot Learning, Fall 2024

Imitation Learning



https://en.wikipedia.org/wiki/Imitation

Instructor: Guanya Shi

Learning and Control for Agile Robotics Lab

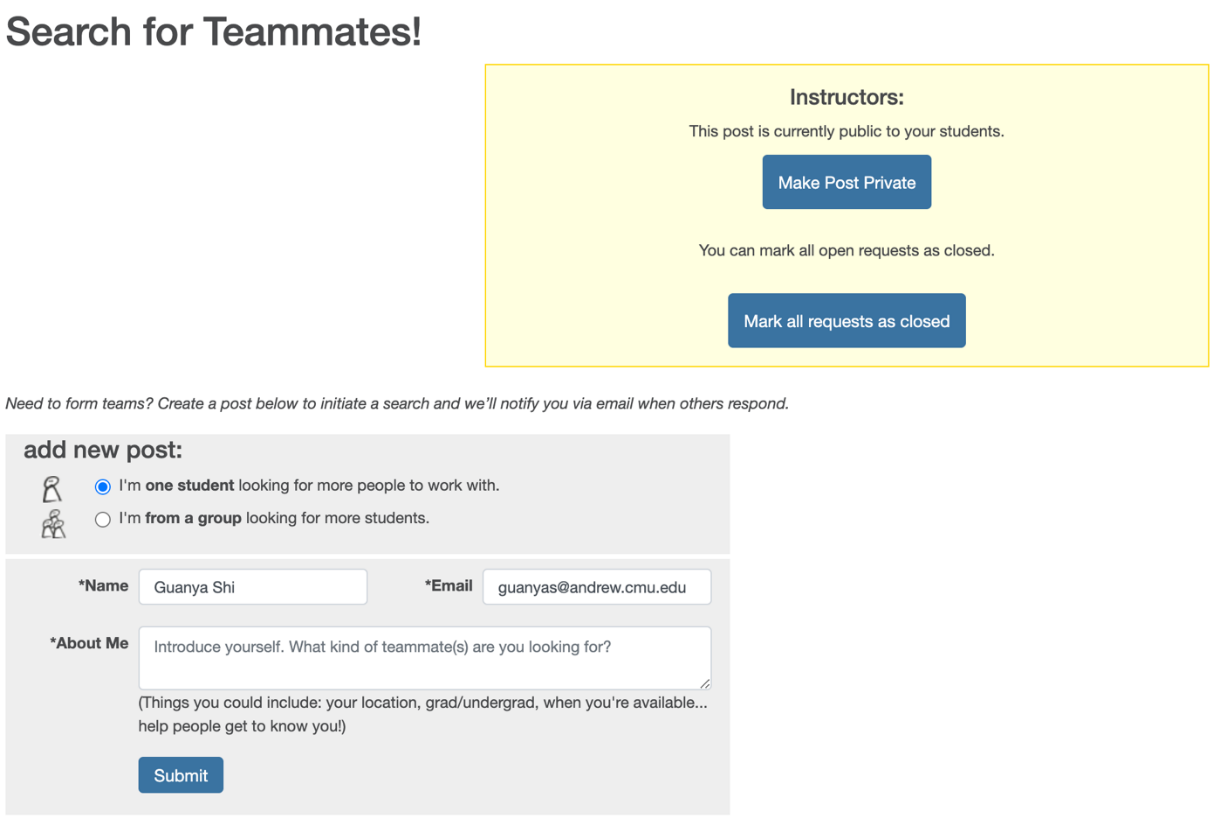
Carnegie Mellon University
The Robotics Institute

Recording Reminder

- This session will be audio recorded for educational use by other students in this course.

Project

- ❑ Please carefully read the Piazza post!
- ❑ Start thinking about your project and searching for teammates!
  - Project proposal deadline: Oct 11
- ❑ AWS credit request form: deadline this Friday (Sep 13)



Course Structure

Latest schedule: https://16-831.github.io/fall24/lectures

- Topic group 1: Course introduction: What is robot learning and why?
- Topic group 2: Machine/Deep learning refresher
- Topic group 3: Imitation learning
- Topic group 4: Model-free reinforcement learning
- Topic group 5: Model-based reinforcement learning
- Topic group 6: Bandit, preference-based learning, exploration
- Topic group 7: Reinforcement learning from offline data (one guest lecture)
- Topic group 8: Specialized topics (one guest lecture)
- Project presentation

Outline

- ❑ Notations, MDP, POMDP
- ❑ What is imitation learning?
- ❑ Why/how is imitation learning different from standard supervised learning?
    - Does behavior cloning (BC) work?
- ❑ How to address the issue?
- ❑ Deep imitation learning via generative models

(Some slides are from 22F, 23F, Berkeley CS 285, and CMU 10-703)

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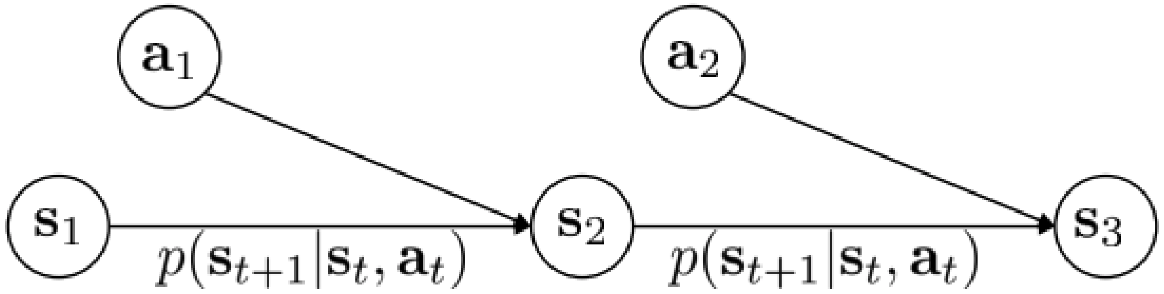
MDP

- Definitions:
  - S is the state space. $s_t \in S$ is the state at time step t.
  - A is the action space. $a_t \in A$ is the action at time step t.
  - O is the observation space. $o_t \in O$ is the observation at time step t.
  - p is the one-step transition probability (a.k.a., dynamics): $s_{t+1} \sim p(\cdot \mid s_t, a_t)$
  - h is the observation model: $o_t \sim h(\cdot \mid s_t)$
  - $r: S \times A \to \mathbb{R}$ is the reward function.

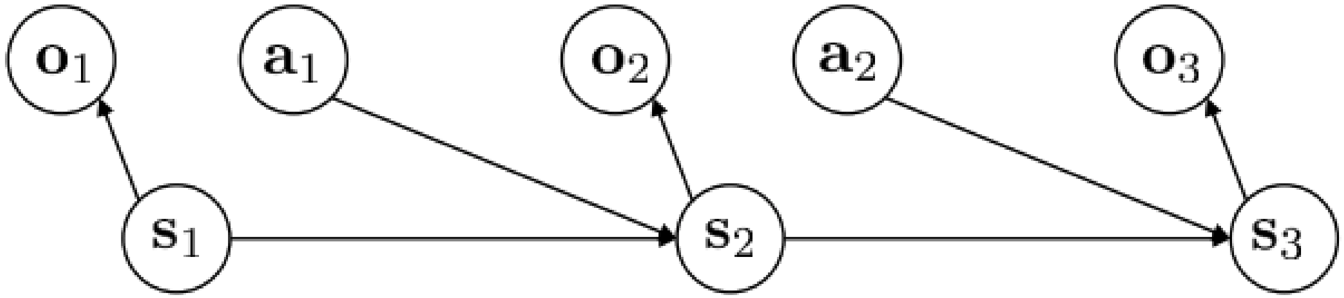
- A Markov decision process (MDP) is a tuple (S, A, p, r)
  - Goal: learn a policy $\pi_\theta(a_t|s_t)$



POMDP

- Definitions:
  - S is the state space. $s_t \in S$ is the state at time step t.
  - A is the action space. $a_t \in A$ is the action at time step t.
  - O is the observation space. $o_t \in O$ is the observation at time step t.
  - p is the one-step transition probability (a.k.a., dynamics): $s_{t+1} \sim p(\cdot \mid s_t, a_t)$
  - h is the observation model: $o_t \sim h(\cdot \mid s_t)$
  - $r: S \times A \rightarrow \mathbb{R}$ is the reward function.

- A partially observed Markov decision process (POMDP) is a tuple (S, A, O, p, h, r)
  - Goal: learn a policy $\pi_\theta(a_t|\text{observations})$



Markov Property

- Definitions:
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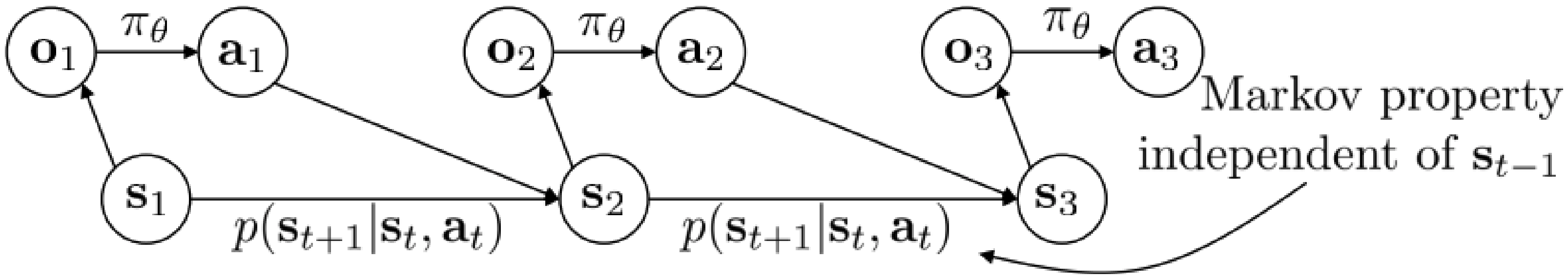
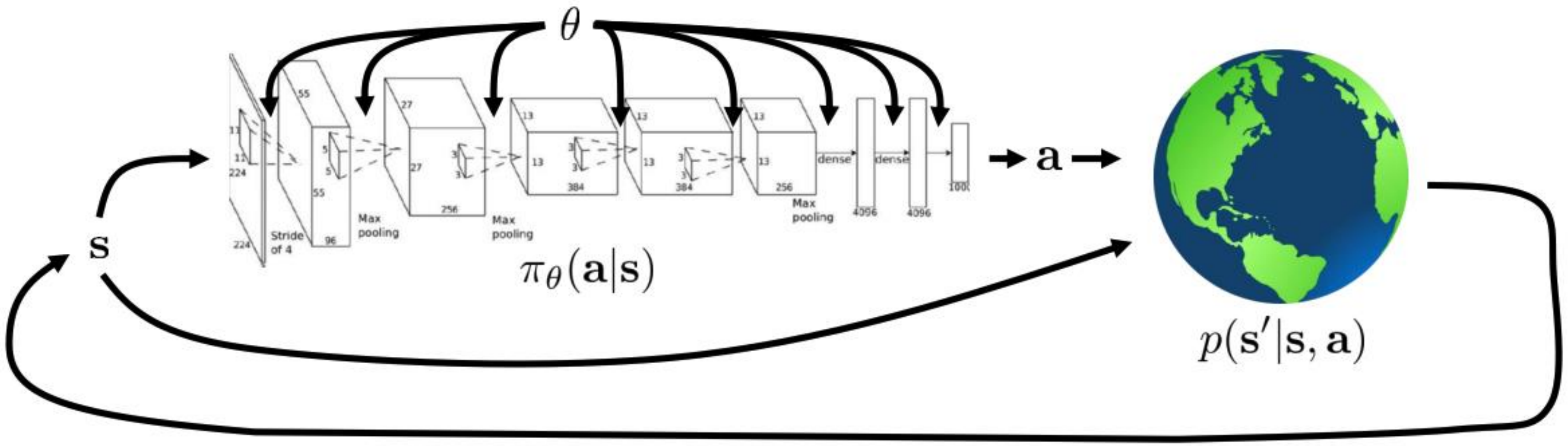
- Why Markov?
  - Allow us to throw away history once state is known!



Andrey Markov

The Goal of Reinforcement Learning and Control

- Finite horizon case: T is finite
- Infinite horizon case: $T = \infty$
- The cumulative reward can be discounted: $\sum_t \gamma^t r(s_t, a_t)$ where $0 < \gamma \leq 1$



$$\underbrace{p_\theta(\mathbf{s}_1, \mathbf{a}_1, \ldots, \mathbf{s}_T, \mathbf{a}_T)}_{\pi_\theta(\tau)} = p(\mathbf{s}_1) \prod_{t=1}^{T} \pi_\theta(\mathbf{a}_t|\mathbf{s}_t) p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$$

$$\theta^\star = \arg\max_\theta E_{\tau \sim p_\theta(\tau)} \left[\sum_t r(\mathbf{s}_t, \mathbf{a}_t) \right]$$

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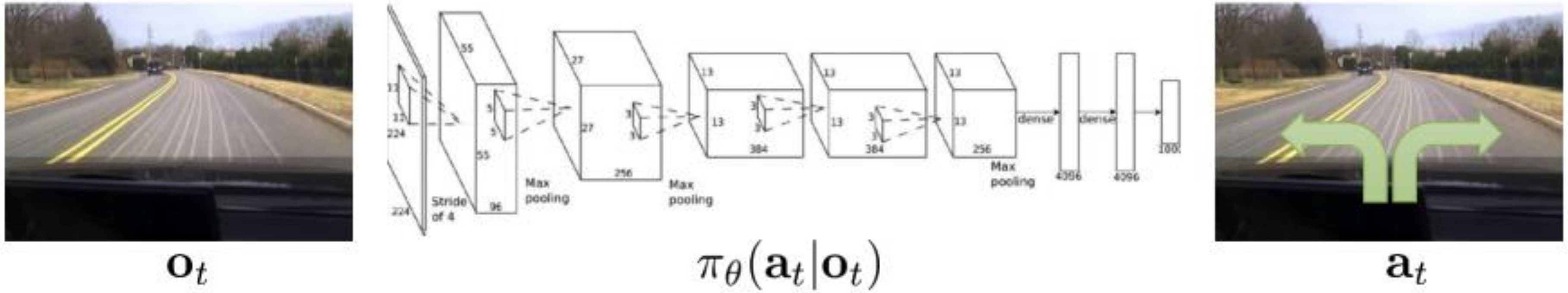
(Some slides are from 22F, 23F, Berkeley CS 285, and CMU 10-703)

Imitation Learning

❑ Main idea:

- Collect expert data (observation/state and action pairs)
- Train a function to map observations/states to actions



Deep Imitation Learning in 1989

- ❑ A CMU paper!
  - CMU has incubated many self-driving companies

**ALVINN:
AN AUTONOMOUS LAND VEHICLE IN A
NEURAL NETWORK**

Dean A. Pomerleau
Computer Science Department
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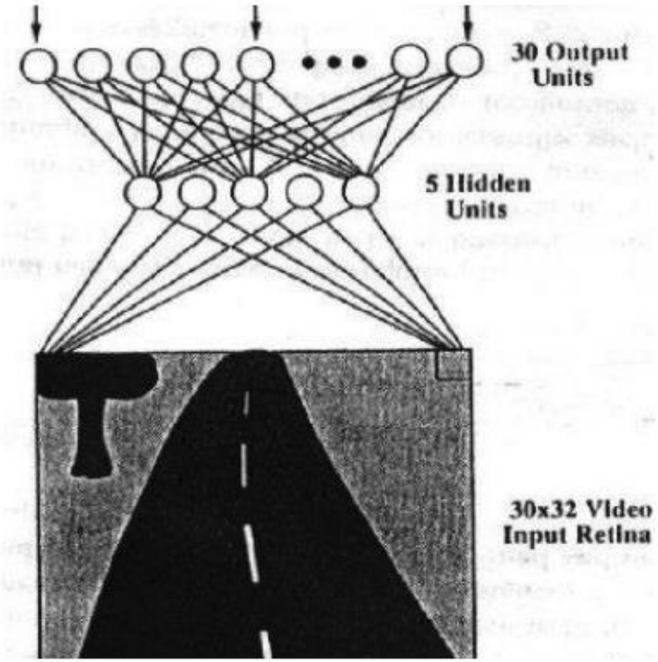
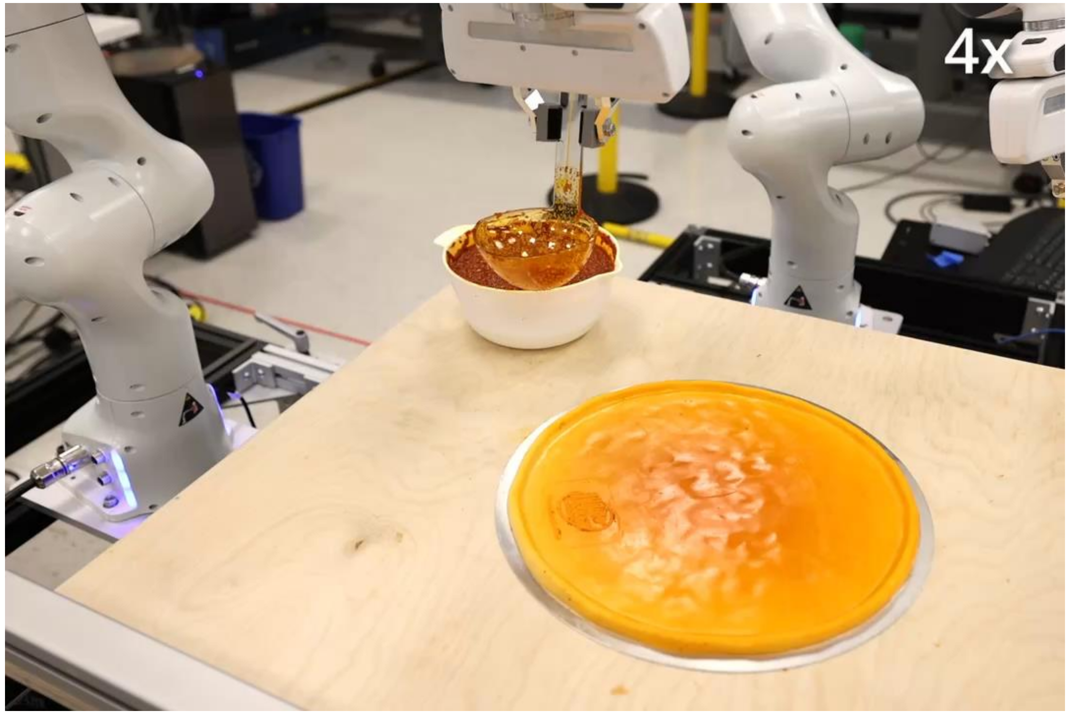
Pittsburgh, PA 15213



Deep Imitation Learning Today

- Left: Mobile ALOHA; Right: Diffusion policy.



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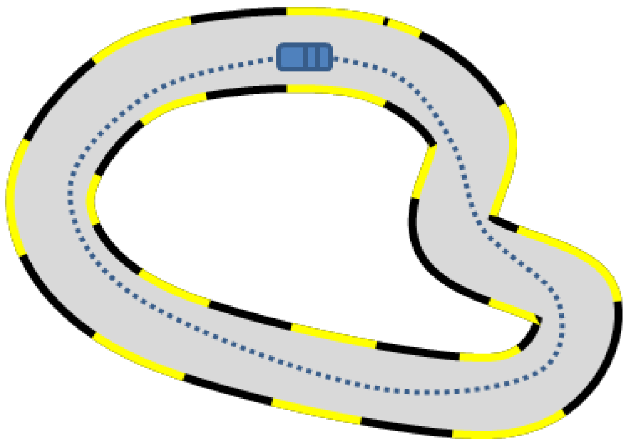
It Sounds Like Supervised Learning …

- Main idea:
  - Collect expert data (observation/state and action pairs)
  - Train a function to map observations/states to actions
  - A.k.a. "**behavior cloning (BC)**"

Expert Trajectories



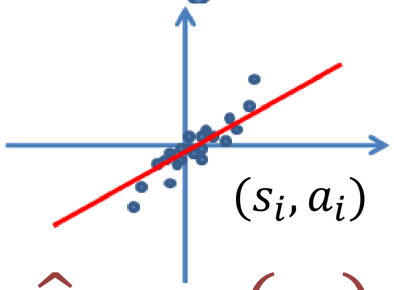
s: state / observation

a: expert action

$$s_0, a_0, s_1, a_1, \dots, s_T, a_T$$

Learned Policy π

Mapping from state (image) to control (steering direction)



$$\hat{a} = \pi(s_i)$$

Supervised Learning

It Sounds Like Supervised Learning …

- High-dimensional observations:

[Pomerleau89, Saxena05, Ross11a]

Input:

Camera Image

Policy

Output:

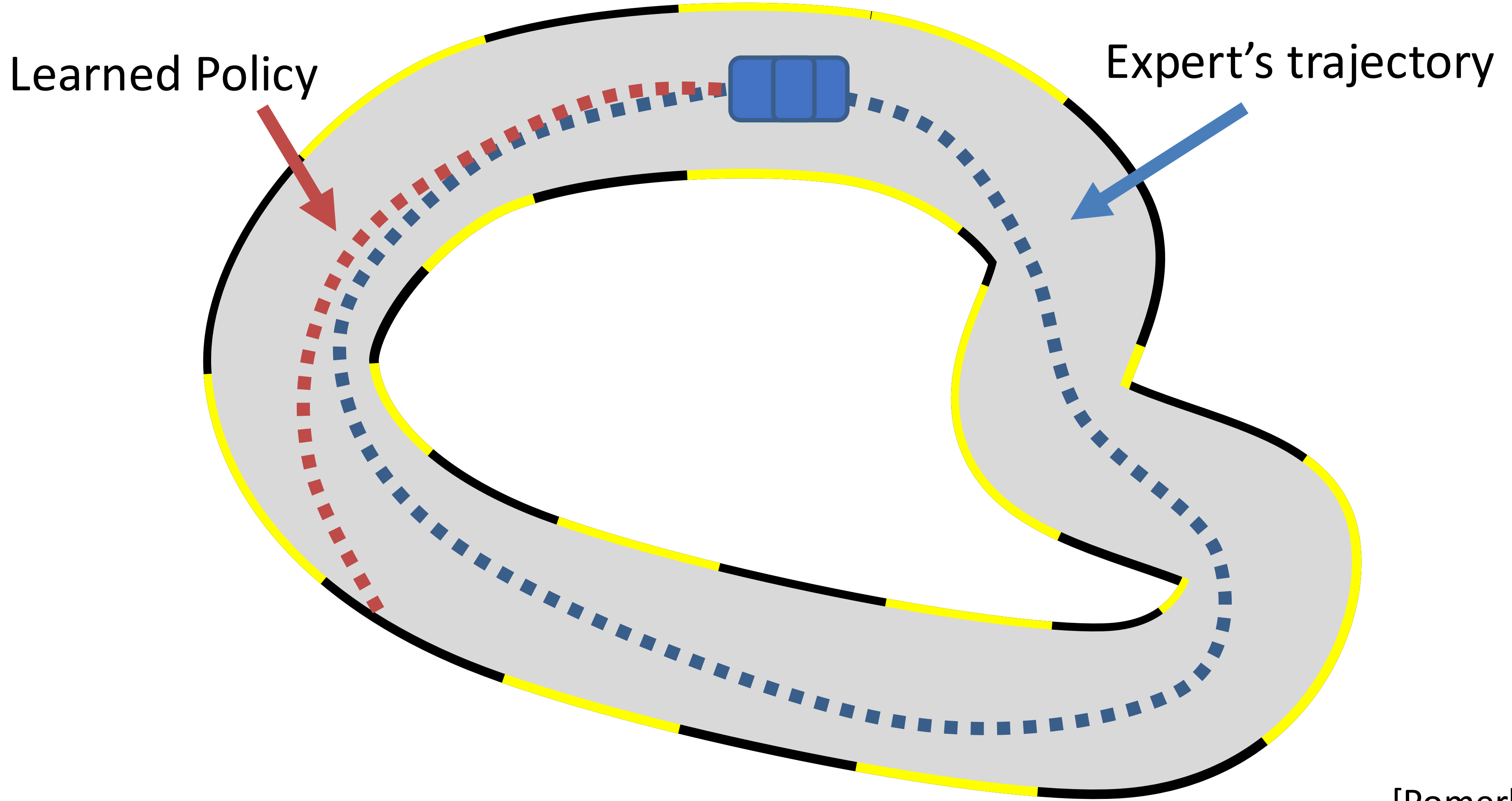
Steering Angle
in [-1, 1]

It Sounds Like Supervised Learning …

- Does BC work?

It Is NOT Standard Supervised Learning!

- ❑ Behavior cloning (BC) does work.
- ❑ What goes wrong?



[Pomerleau89,Daume09]

It Is NOT Standard Supervised Learning!

- The i.i.d. (independent and identically distributed) assumption doesn't hold!
- Recap the supervised learning story (lecture 3):

Theorem (Vapnik and Chervonenkis, 1971)

Consider $\mathcal{F} \subseteq \{-1,1\}^{\mathcal{X}}$, $\mathcal{Y} = \{\pm 1\}$, $\ell = \ell_{01}$.
For every prob distribution P on $\mathcal{X} \times \{-1,1\}$,
with probability $1-\delta$ over n iid examples $(x_1, y_1), \ldots, (x_n, y_n)$,
every f in $\mathcal{F}$ satisfies

"train error"

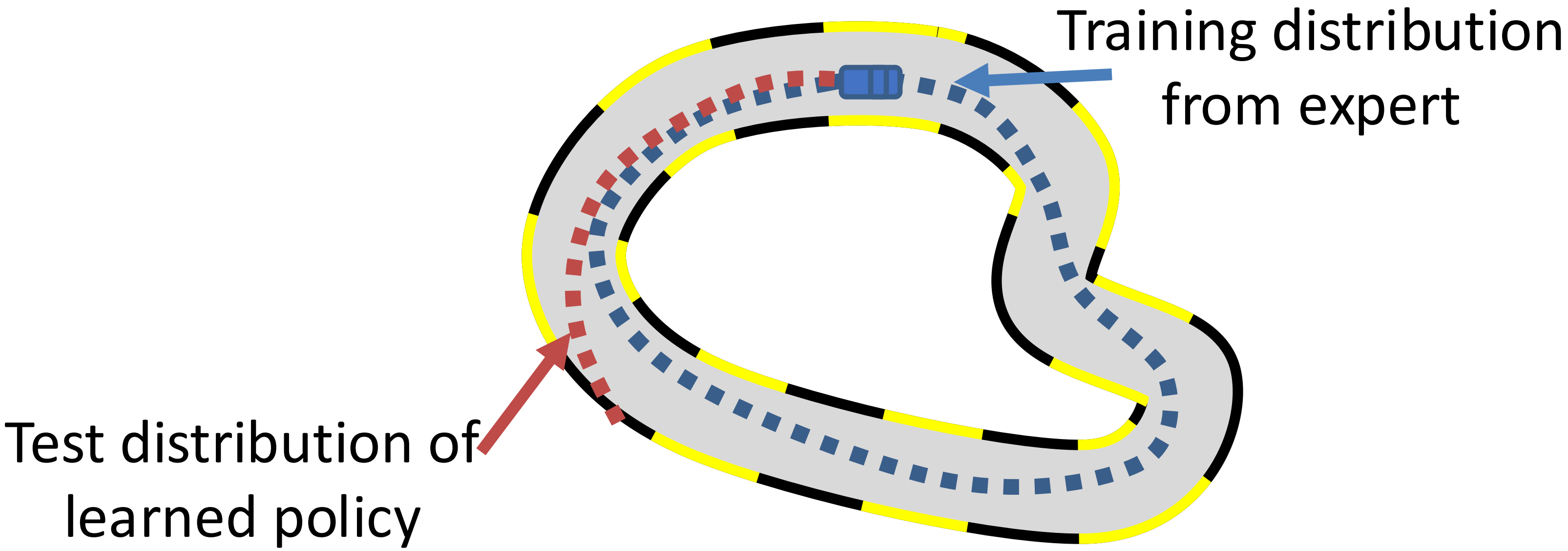
"test error"

$$\mathbb{E}\ell_f \leq \hat{\mathbb{E}}\ell_f + O\left(\sqrt{\frac{d_{VC}(\mathcal{F})}{n}}\right).$$

- (x_i, y_i) is i.i.d. sampled from some distribution
- Basically, "low training error" -> "good test performance"

It Is NOT Standard Supervised Learning!

- ❑ The expert data only contains "good" behaviors
- ❑ Therefore, a small mistake in the test time will cause cascading failures

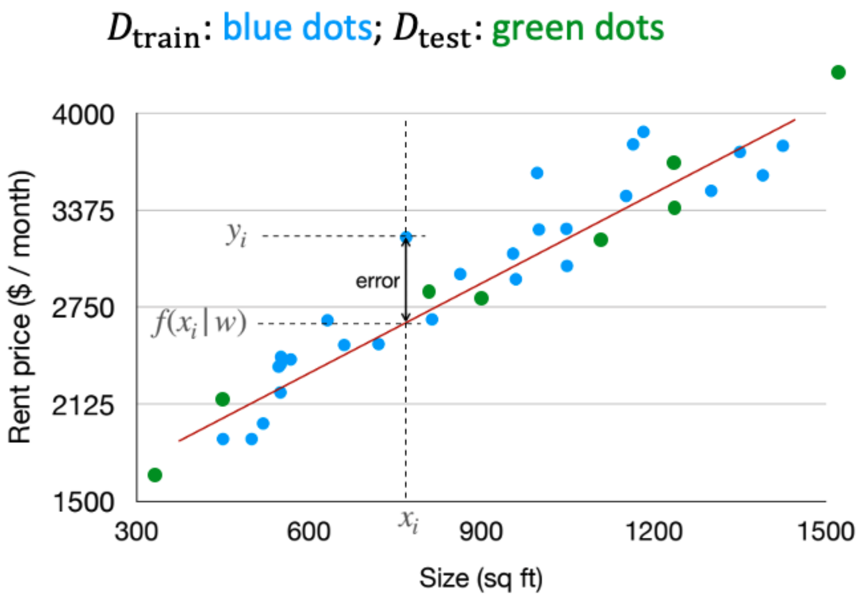


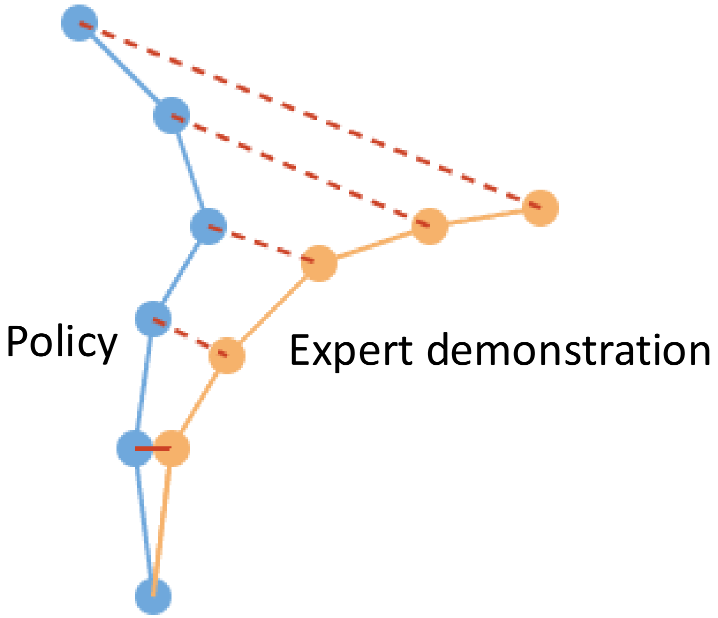
Low Training Error

Good Test Performance

It Is NOT Standard Supervised Learning!

❑ Recap the **domain shift** problem in ML/DL (lecture 3):





❑ Even worse in sequential decision making! The domain shift is “**dynamic**” and “**fatal**”

- Policy only trained with expert data
- Expert has almost zero chance seeing “bad” state
- Statistically speaking, imitation learning is **cliff walking**
- Once we make one mistake, we are outside of the training distribution

Some Analysis

- ❑ Consider a "cliff walking" problem
  - Test error quadratic in # timesteps!

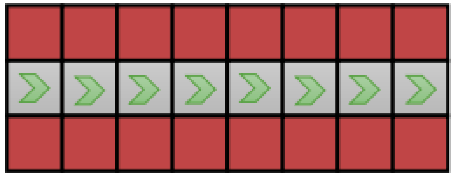
$$c(\mathbf{s},\mathbf{a}) = \begin{cases} 0 \text{ if } \mathbf{a} = \pi^\star(\mathbf{s}) \\ 1 \text{ otherwise} \end{cases}$$

$$E\left[\sum_t c(\mathbf{s}_t,\mathbf{a}_t)\right] \le \epsilon T + \underbrace{\qquad\qquad\qquad\qquad\qquad\qquad}_{T \text{ terms, each } O(\epsilon T)}$$

$$O(\epsilon T^2)$$

assume: $\pi_\theta(\mathbf{a} \neq \pi^\star(\mathbf{s})|\mathbf{s}) \le \epsilon$

for all $\mathbf{s} \in \mathcal{D}_{\text{train}}$



- ❑ Why is it so pessimistic?
  - In reality, we can often **recover** from mistakes
  - But this **paradox** still exists: IL could work better if the data has more mistakes!

For more analysis, see Ross et al. "A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning"

This Issue is Also in Sequential Prediction

Ground Truth

Canonical Approach

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How to Address this Problem?

- ❑ Be smart about how we collect (and augment) our data
- ❑ Change the algorithm (DAgger)
- ❑ Next time: use very powerful models that make very few mistakes
- ❑ Lecture 23: use multi-task learning

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ALVINN:
AN AUTONOMOUS LAND VEHICLE IN A NEURAL NETWORK

Dean A. Pomerleau
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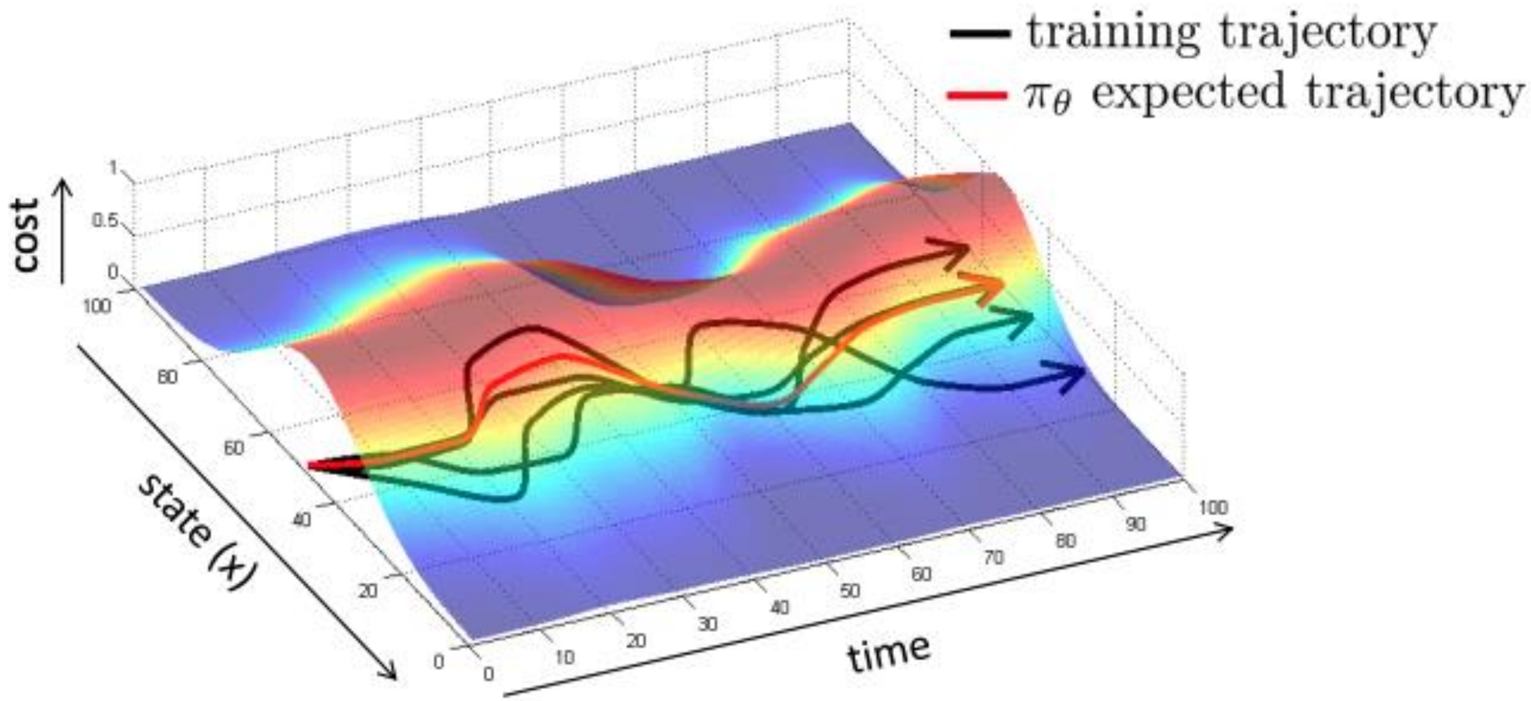
"...the network must not solely be shown examples of accurate driving, but also how to recover (i.e. return to the road center) once a mistake has been made."

What Makes BC Easy/Hard?

- ❑ Be smart about how we collect (and augment) our data
- ❑ Idea 1: Intentionally add mistakes and corrections
  - The mistakes hurt, but the corrections help, often more than the mistakes hurt
- ❑ Idea 2: Use data augmentation
  - Add some “fake” or “synthetic” data that illustrates corrections

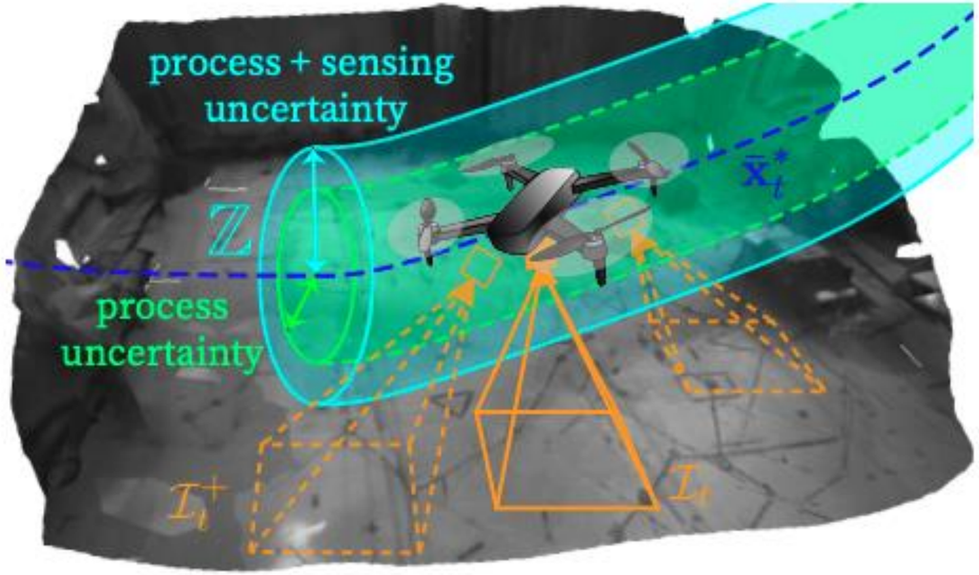


Data Augmentation Example

- ❑ Augment an MPC expert using a tube

Demonstration-Efficient Guided Policy Search via Imitation of Robust Tube MPC

Andrea Tagliabue, Dong-Ki Kim, Michael Everett, Jonathan P. How

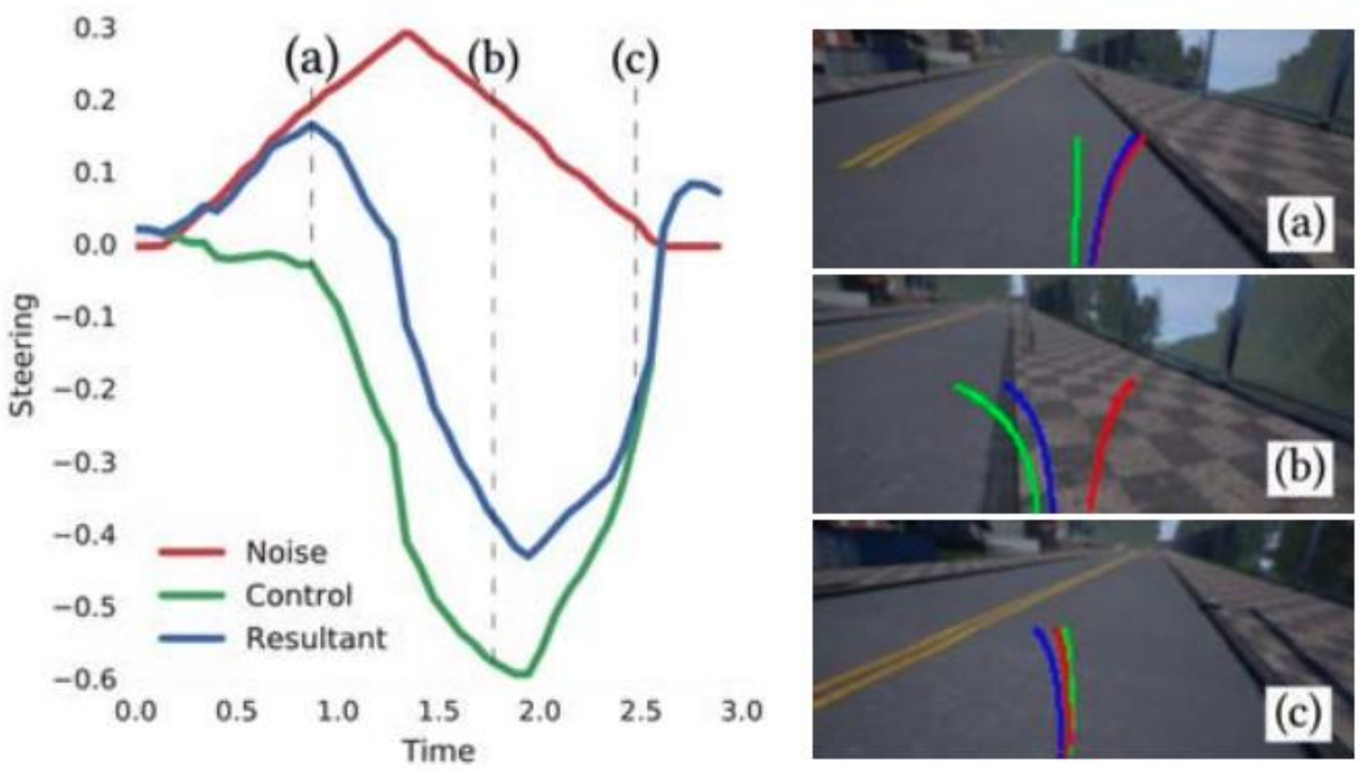


Data Augmentation Example

- ❑ Perturb the driver's steering signal

End-to-end Driving via Conditional Imitation Learning

Felipe Codevilla[1,2] Matthias Müller[1,3] Antonio López[2] Vladlen Koltun[1] Alexey Dosovitskiy[1]



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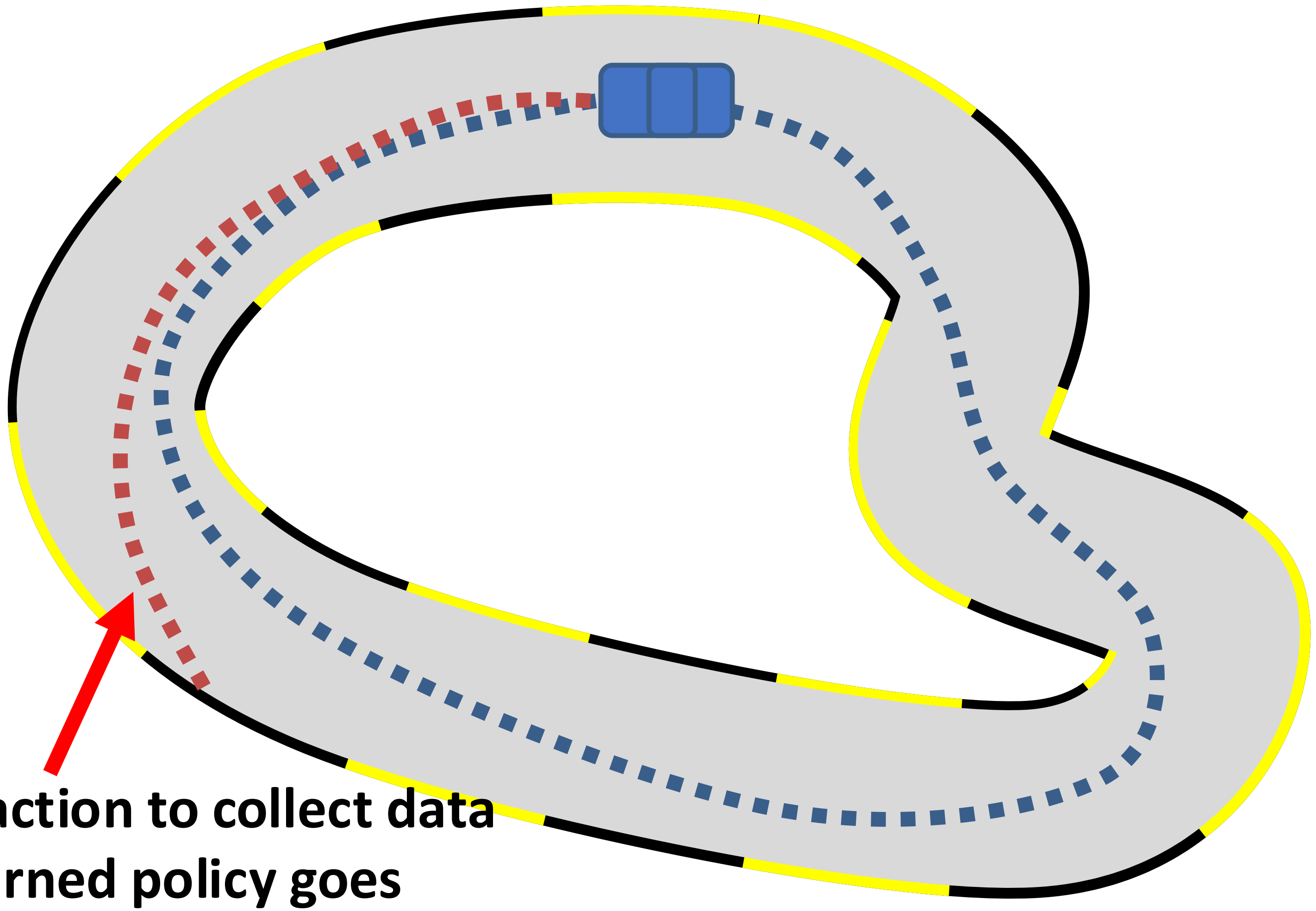
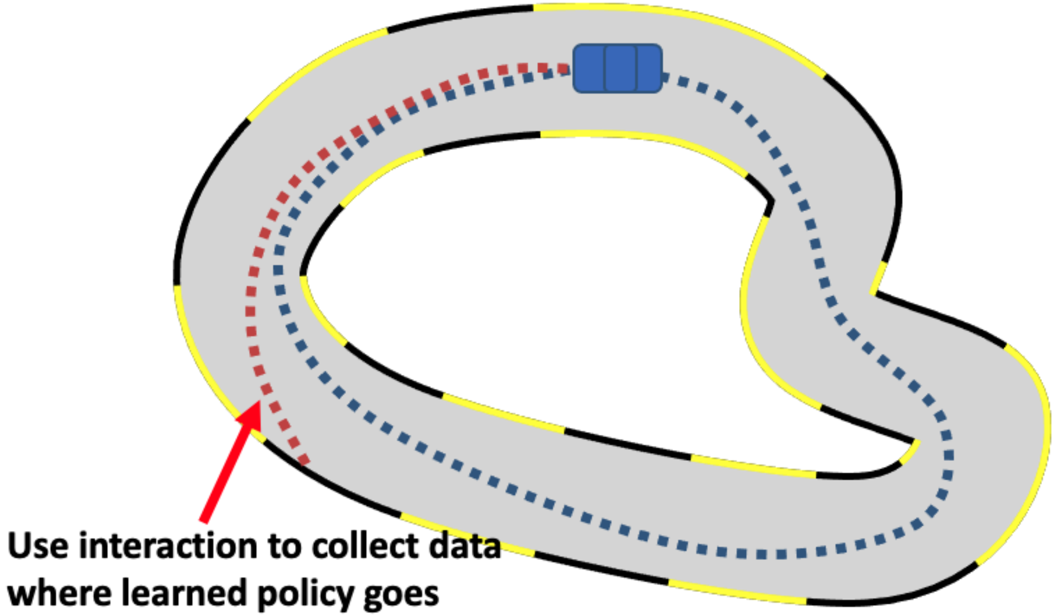
Key Idea: Interaction



Use interaction to collect data where learned policy goes

Key Idea: Interaction

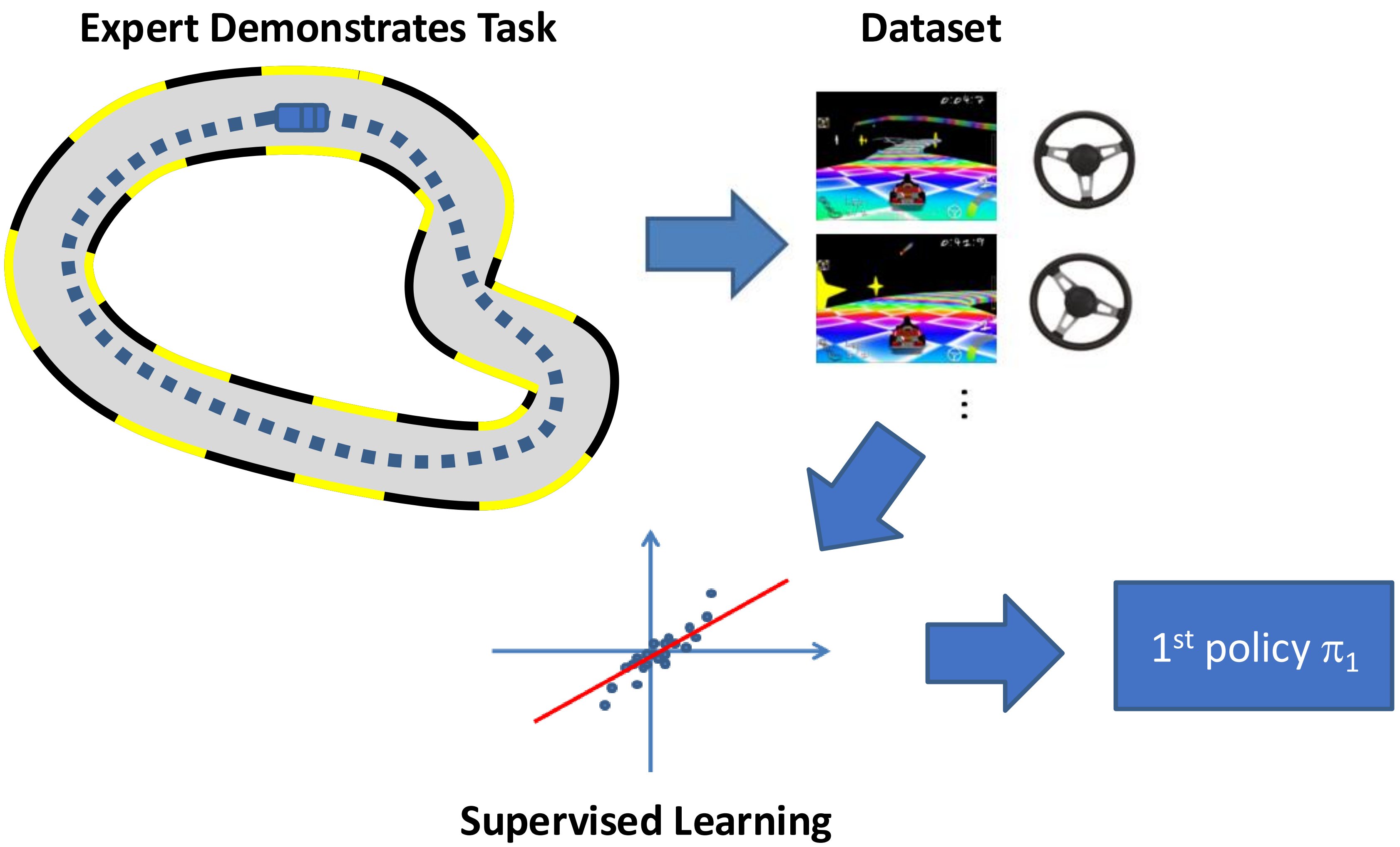


Train policy on expert data

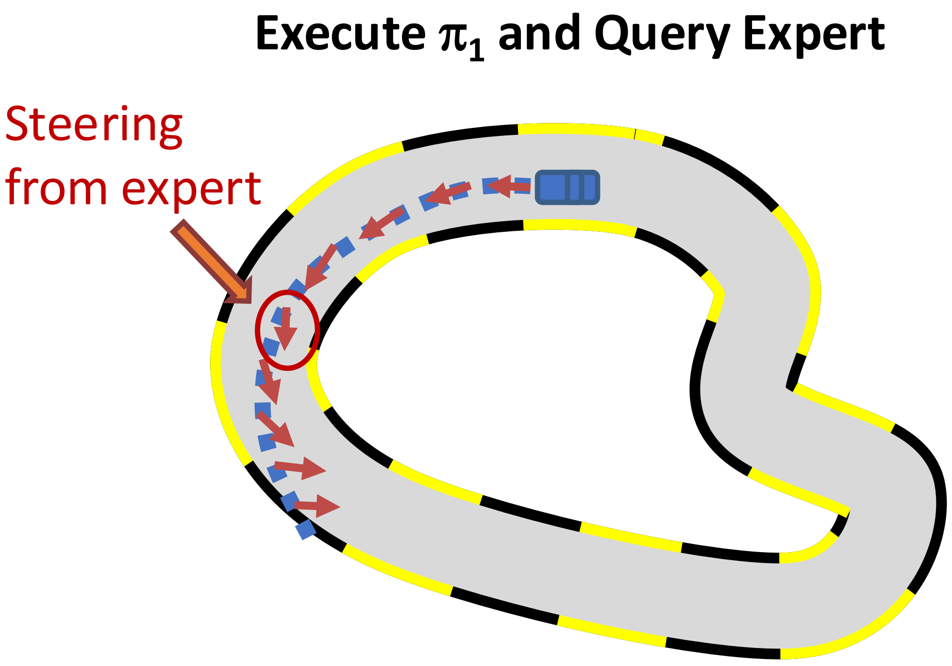
Run policy

Collect new expert data to recover from mistakes

1^{st} iteration



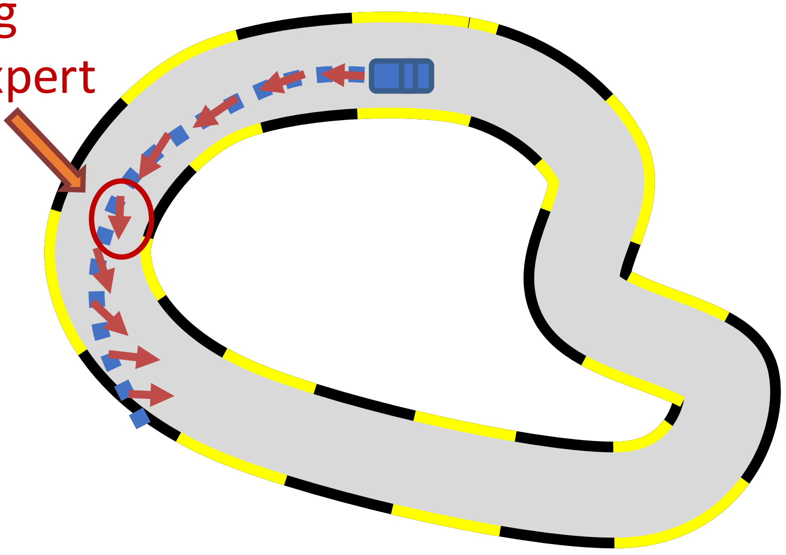
2^{nd} iteration



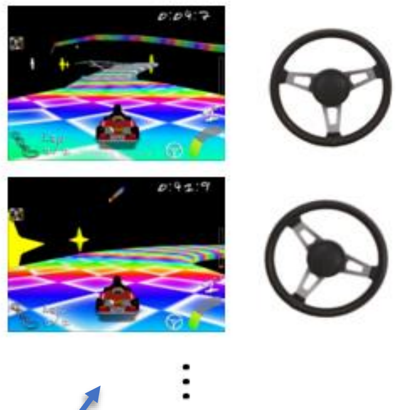
2^{nd} iteration

Execute π_1 and Query Expert

Steering
from expert



New Data



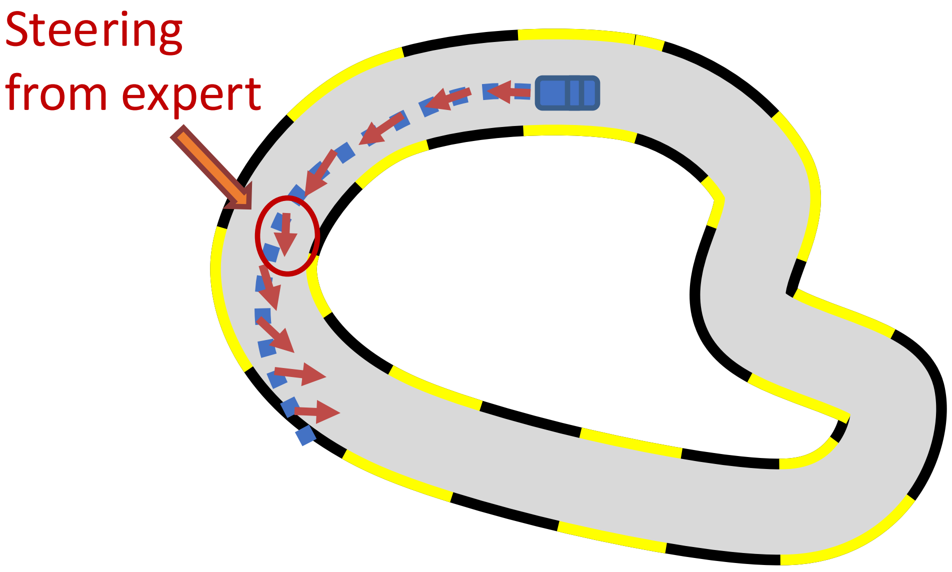
States from
the learned policy

[Ross11a]

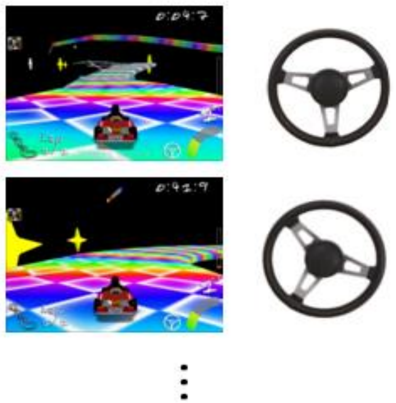
DAgger: Dataset Aggregation

2nd iteration

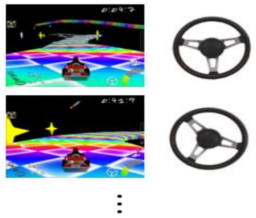
Execute π_1 and Query Expert



New Data



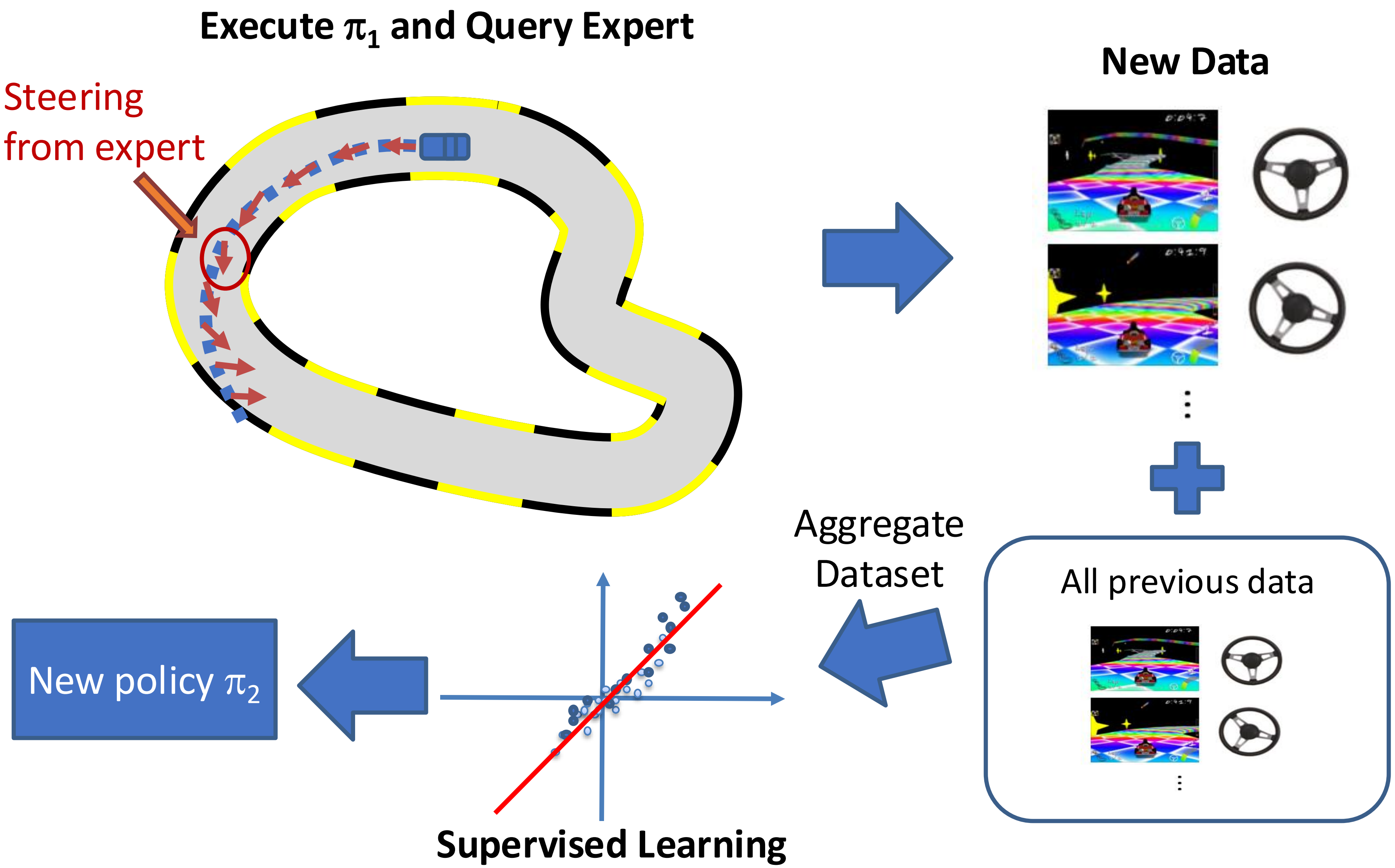
All previous data



[Ross11a]

DAgger: Dataset Aggregation

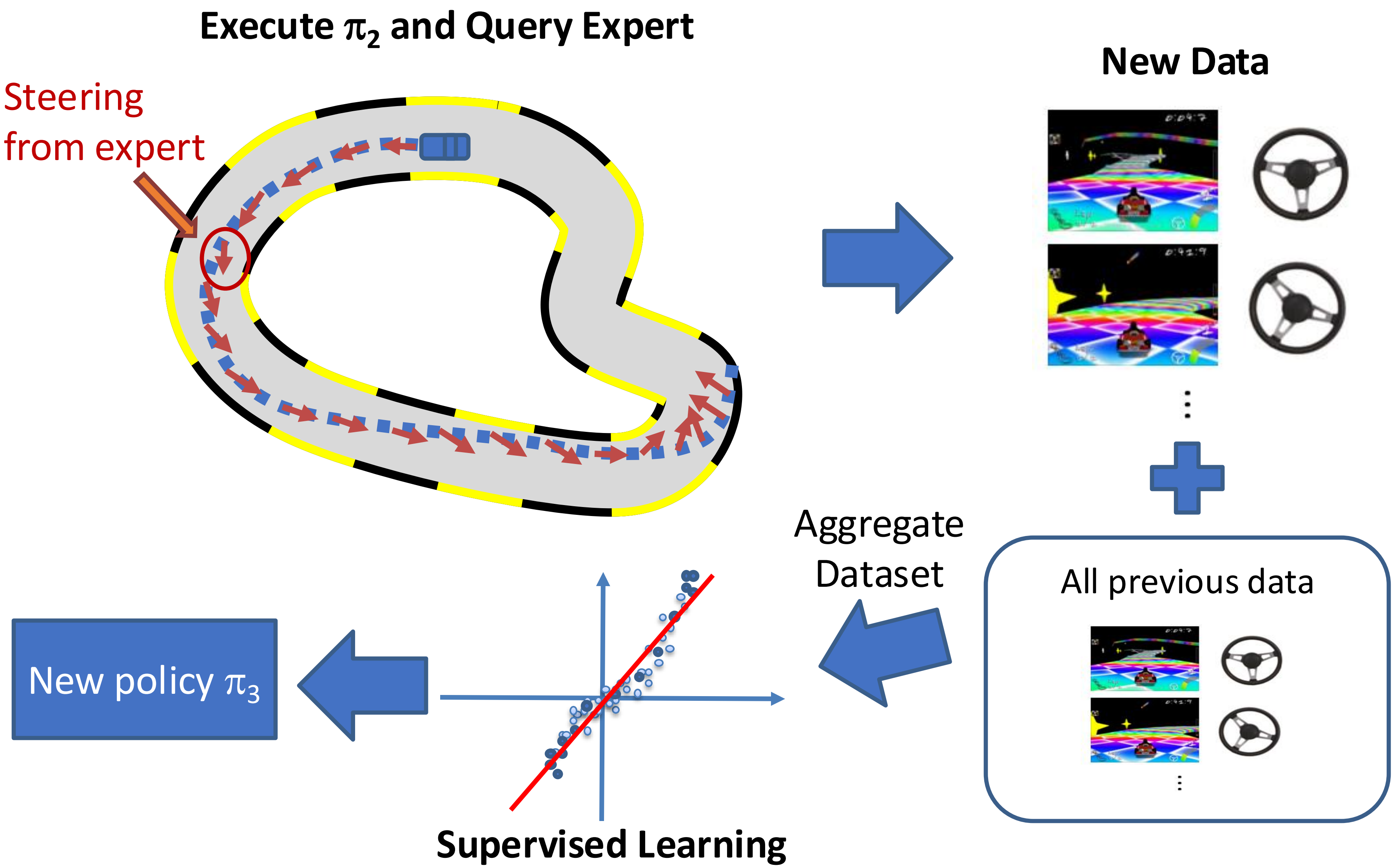
2^{nd} iteration



[Ross11a]

DAgger: Dataset Aggregation

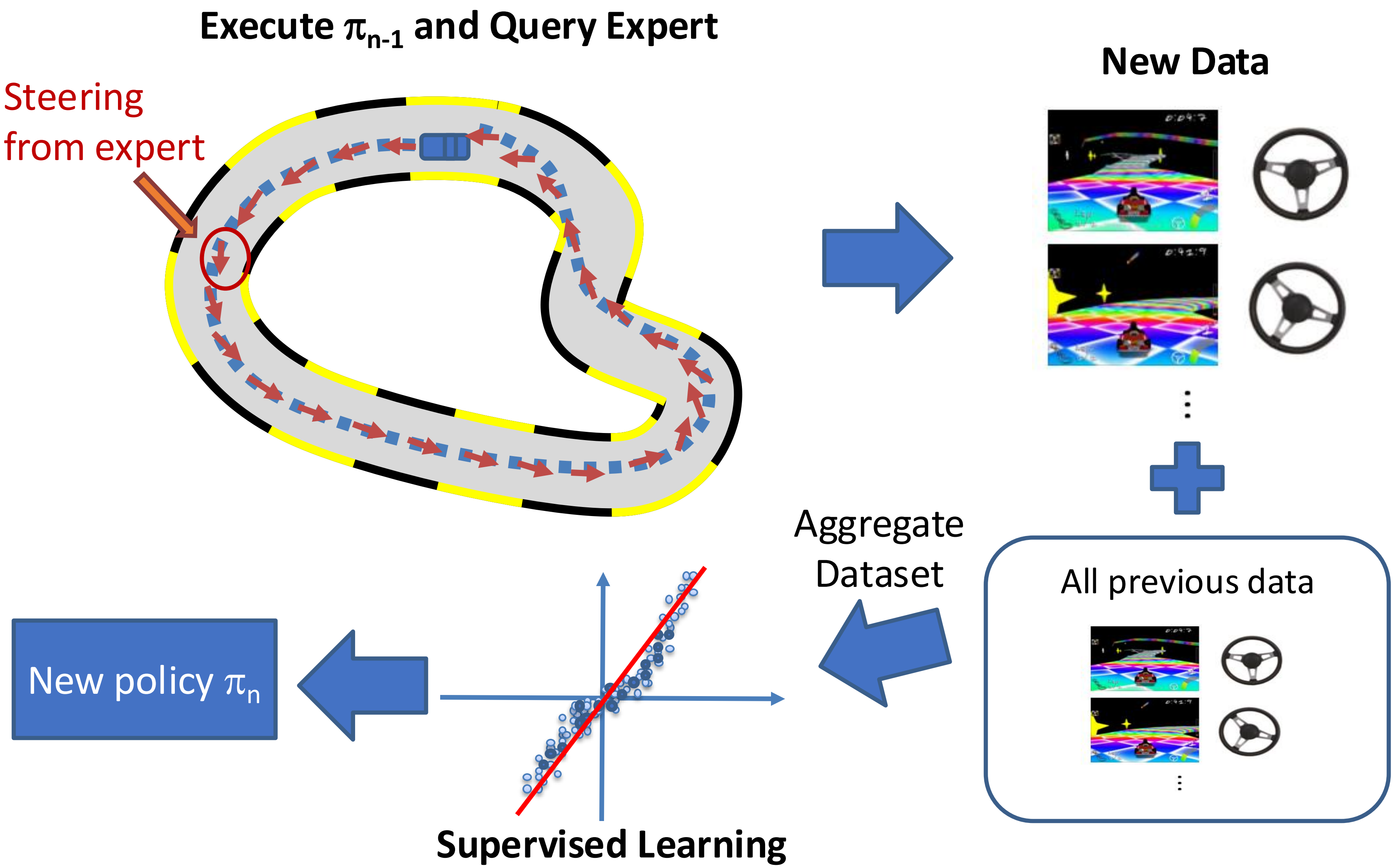
3rd iteration



[Ross11a]

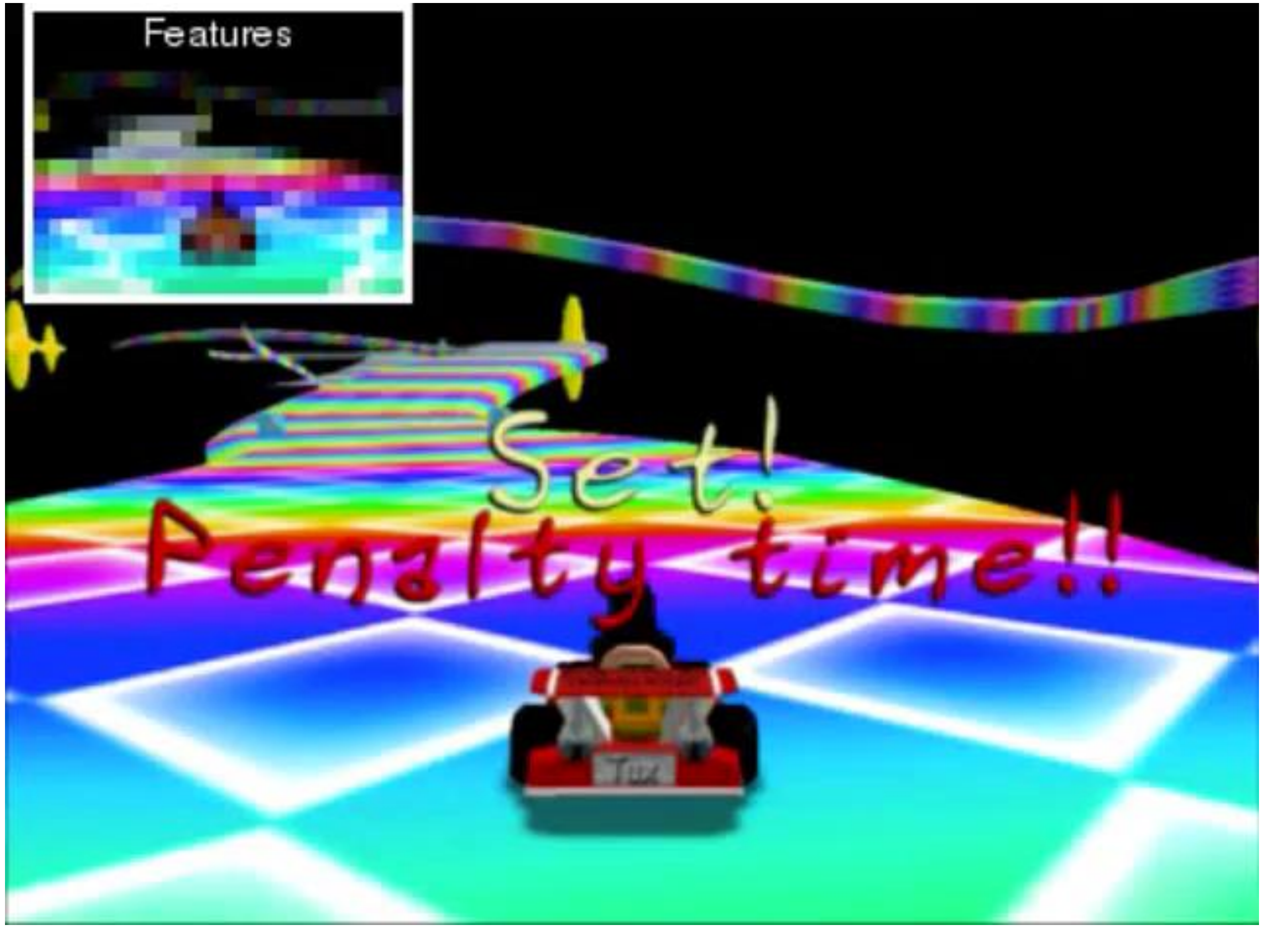
DAgger: Dataset Aggregation

n^{th} iteration



It Works!

- ❑ Left: BC; Right: DAgger



Summary of DAgger

- ❑ From CMU (2011)!
- ❑ It is essentially an online learning algorithm
  - With regret guarantees!
- ❑ Problem: need to keep querying the expert

1. train $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ from human data $\mathcal{D} = \{\mathbf{o}_1, \mathbf{a}_1, \ldots, \mathbf{o}_N, \mathbf{a}_N\}$
2. run $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ to get dataset $\mathcal{D}_\pi = \{\mathbf{o}_1, \ldots, \mathbf{o}_M\}$
3. Ask human to label $\mathcal{D}_\pi$ with actions $\mathbf{a}_t$
4. Aggregate: $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_\pi$

A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning

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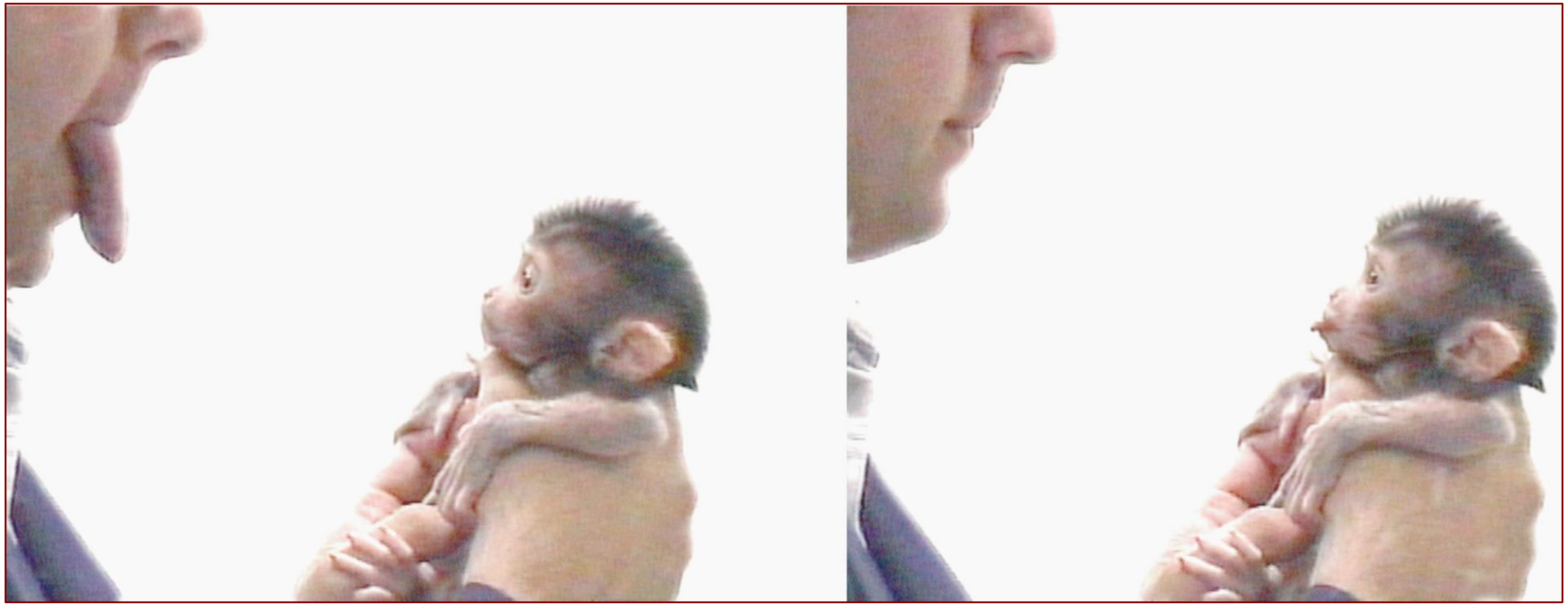
16-831: Introduction to Robot Learning, Fall 2024

Imitation Learning



https://en.wikipedia.org/wiki/Imitation

Instructor: Guanya Shi

Learning and Control for Agile Robotics Lab

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The Robotics Institute

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  - Better data collection or data augmentation
  - Better algorithm: DAgger

Today:

- ❑ Imitation learning with powerful models: using history, multi-modal, …
- ❑ Imitation learning with privileged teachers
- ❑ Deep imitation learning via generative models
  - Very active research area!

(Some slides are from 22F, 23F, Berkeley CS 285, and CMU 10-703)

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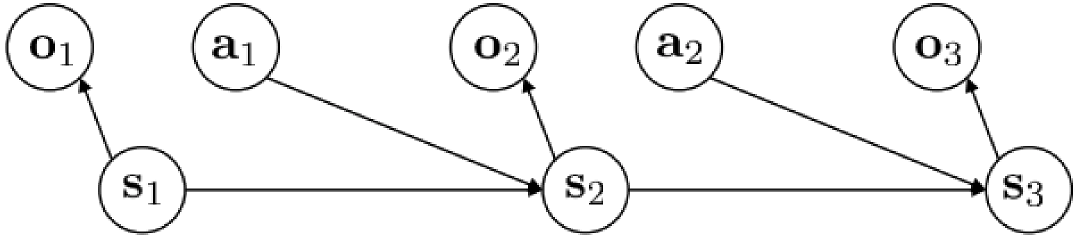
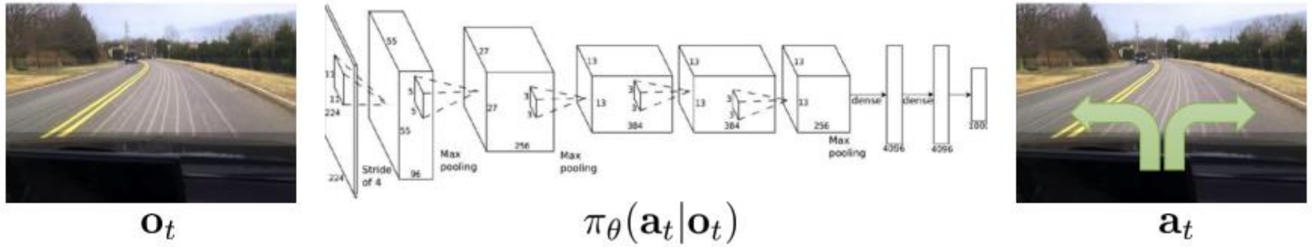
Use History

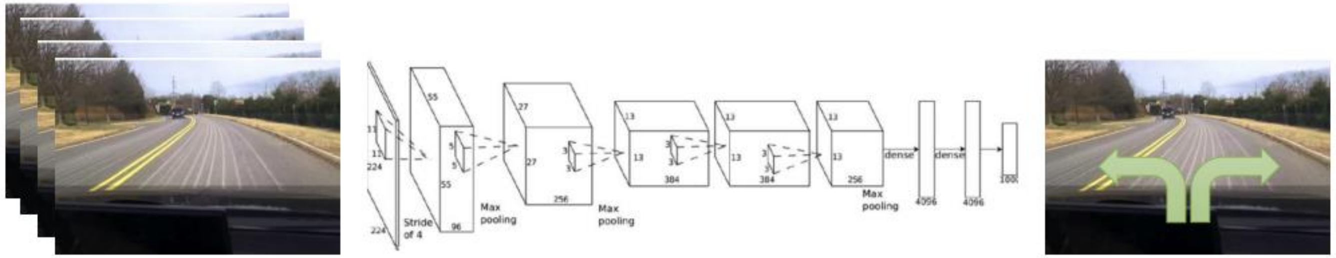
- ❑ Remember: o_t is the observation
- ❑ Idea: learn $\pi_\theta(a_t|o_1, \cdots, o_t)$
  - $\pi_\theta(a_t|o_t)$: If we see the same thing twice, we do the same thing twice, regardless of what happened before – not natural for human demonstrators!
- ❑ Question: does $\pi_\theta(a_t|s_1, \cdots, s_t)$ make sense?
  - From Markovian perspective: not necessarily…
  - From imitation learning perspective: yes!



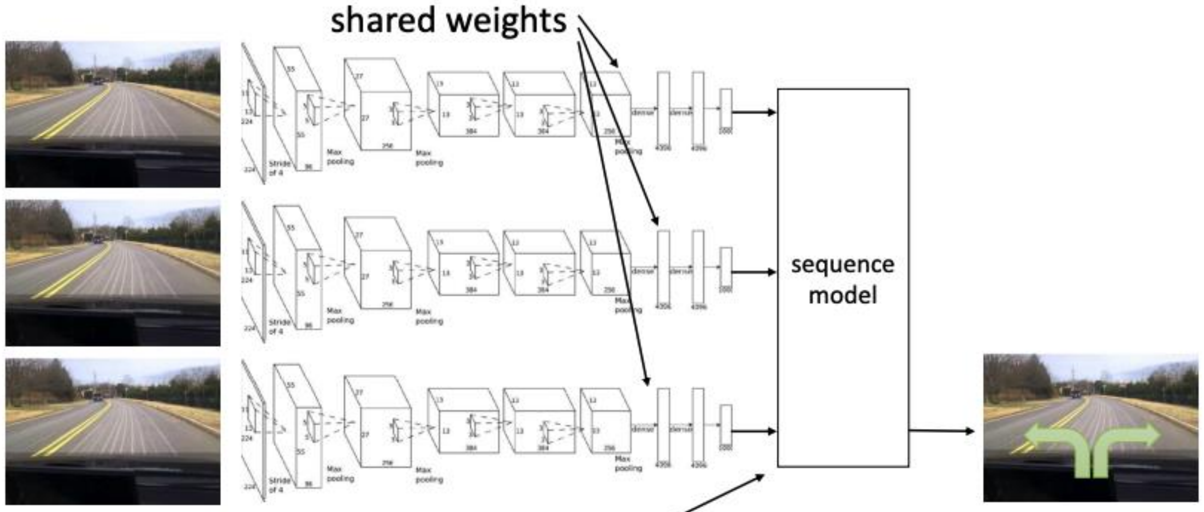


Use History

- ❑ How to represent $\pi_\theta(a_t|o_1, \cdots, o_t)$?
- ❑ "Fully connected"?
  - Variable number of frames & too many weights!



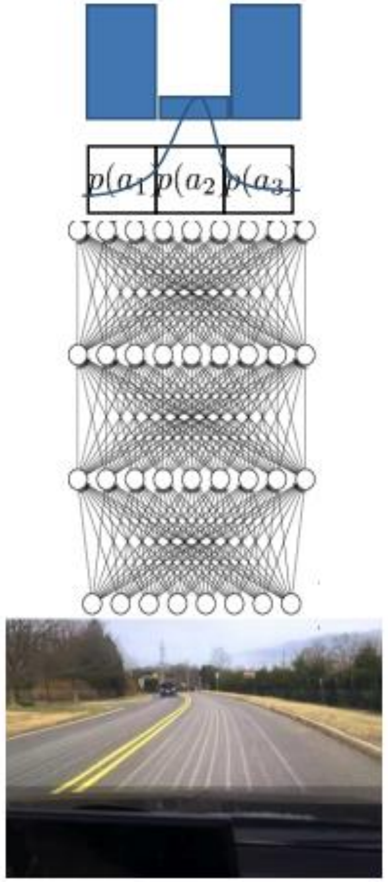
- ❑ Time series models we introduced in lecture 4!

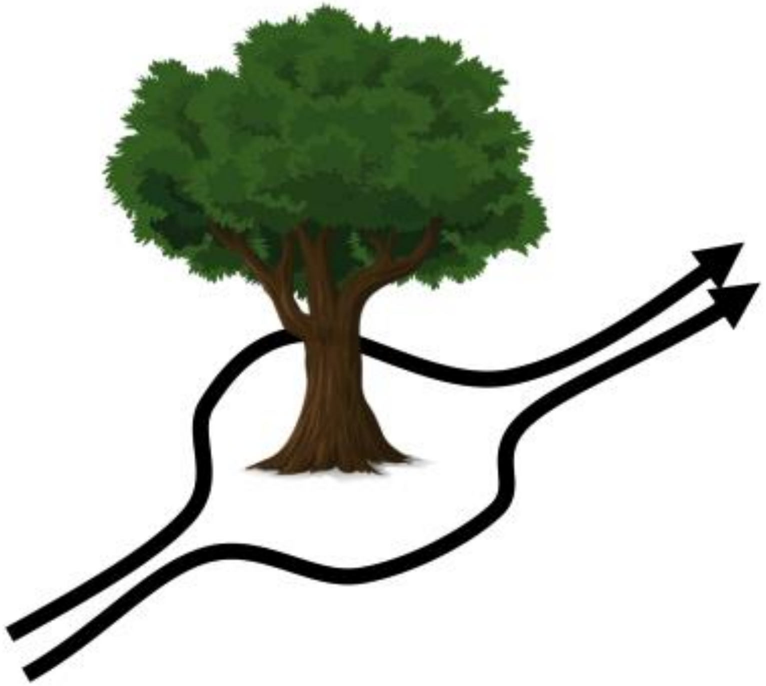


Can be done with Transformers, LSTM cells, etc.

Model Multi-modal Behavior

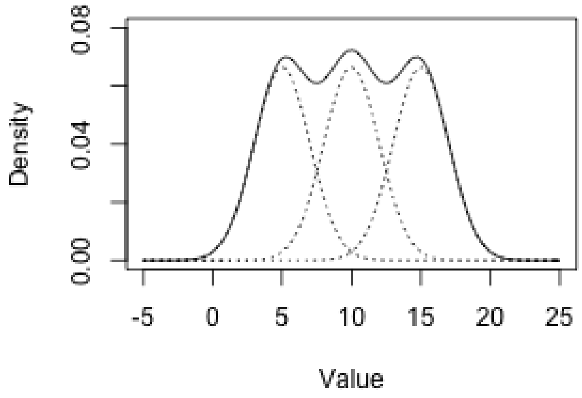
- Multi-modal behavior is common for many demonstrators:
  - Human demonstrator
  - Optimal control
  - Oracle search algorithm

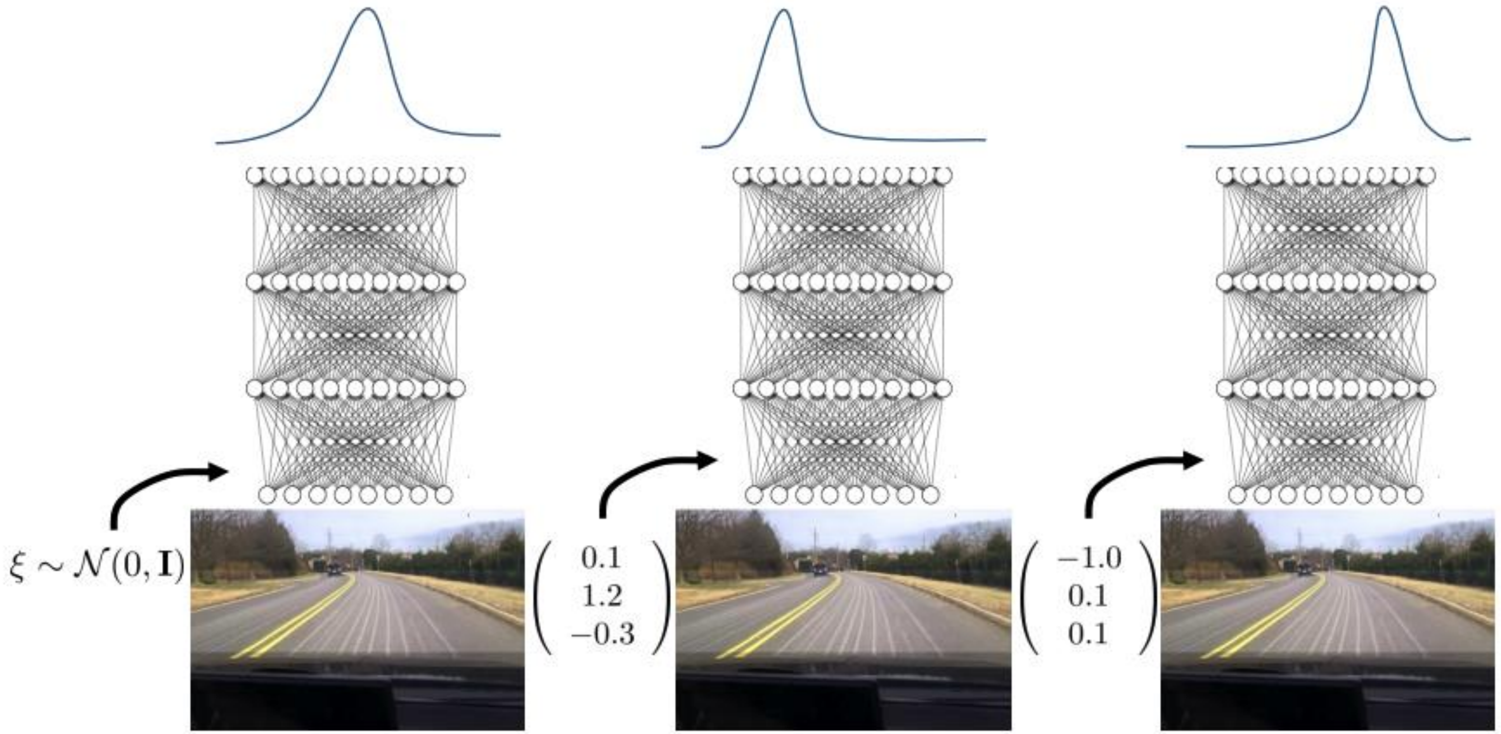




Model Multi-modal Behavior

- How to model multi-modal behavior?
  - Predefine: Gaussian mixture model
  - Latent variable models: (conditional) VAE
  - Diffusion model (more details later)
  - Other generative models (e.g., GAN)





Outline

Last time:

- ❑ Notations, MDP, POMDP
- ❑ What is imitation learning?
- ❑ Why/how is imitation learning different from standard supervised learning?
  - Does behavior cloning (BC) work?
- ❑ How to address the issue?
  - Better data collection or data augmentation
  - Better algorithm: DAgger

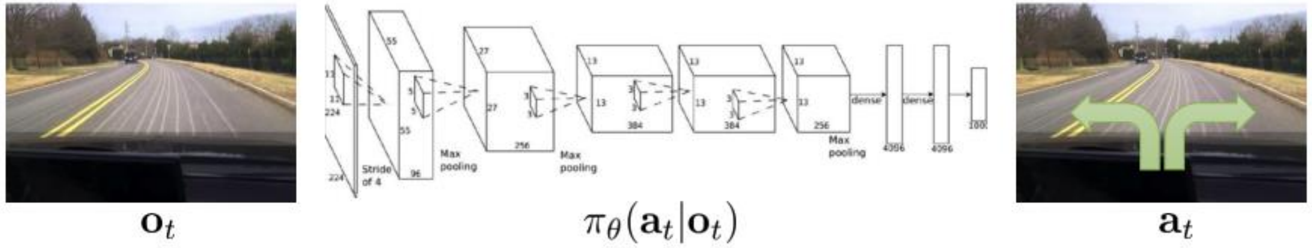
Today:

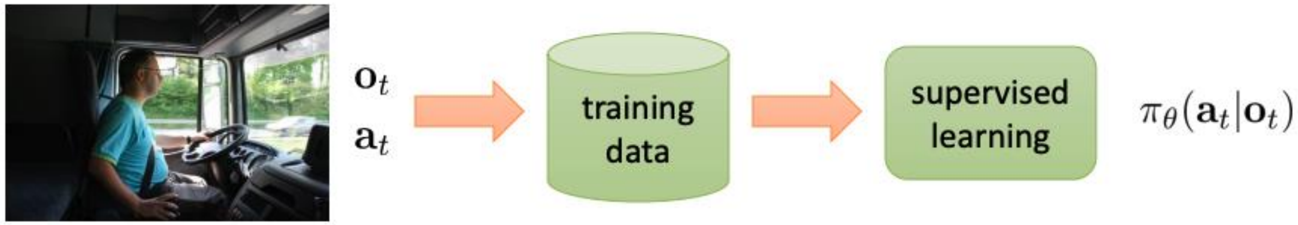
- ❑ Imitation learning with powerful models: using history, multi-modal, …
- ❑ Imitation learning with privileged teachers
- ❑ Deep imitation learning via generative models
  - Very active research area!

(Some slides are from 22F, 23F, Berkeley CS 285, and CMU 10-703)

IL with Privileged Teachers

- ❑ It might be hard to directly learn $\pi_\theta(a_t|o_t)$ (or $\pi_\theta(a_t|o_1, \cdots, o_t)$)
- ❑ o_t is high-dimensional!
- ❑ Idea: obtain a "privileged" teacher $\pi_p(a_t|p_t)$ first
    - p_t contains some "ground truth" information that is not directly available to students
    - Then use $\pi_p(a_t|p_t)$ to generate demonstrations for $\pi_\theta(a_t|o_t)$



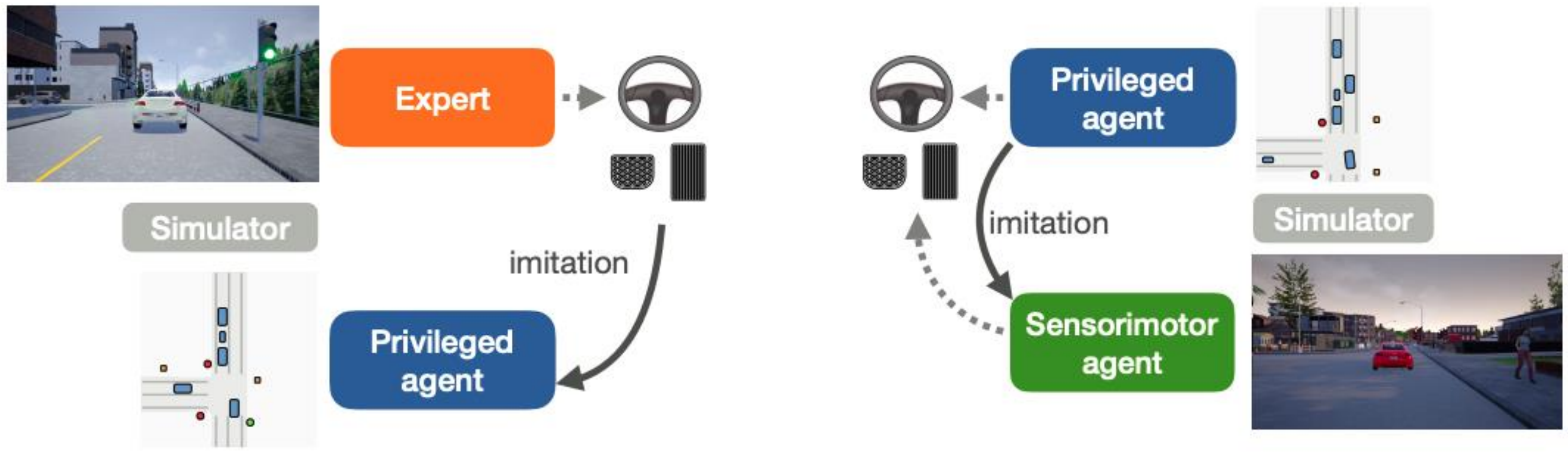


Example of IL with Privileged Teachers: Driving

- ❑ Stage 1: learn a “privileged agent” from expert
    - It knows ground truth state (traffic light, other vehicles’ pos/vel, etc)
- ❑ Stage 2: a sensorimotor student learns from this train privileged agent

Learning by Cheating

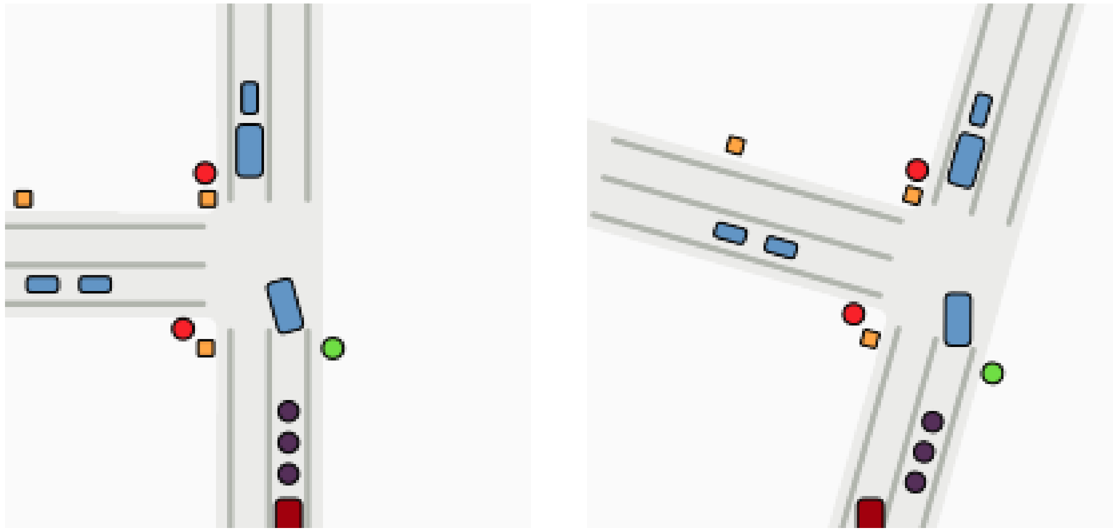
Dian Chen UT Austin
Brady Zhou Intel Labs, UT Austin
Vladlen Koltun Intel Labs
Philipp Krähenbühl UT Austin



(a) Privileged agent imitates the expert (b) Sensorimotor agent imitates the privileged agent

Example of IL with Privileged Teachers: Driving

- ❑ Why it works (compared to direct IL)?
  - The trained "privileged agent" is white-box and can provide as much supervision as you want (e.g., running DAgger)
  - Optimization might be easier
  - Inherently offer some invariance (e.g., night/day driving, different vehicle colors)
  - Easy to do data augmentation



(a) Road map (b) Rotation and shift aug.

Example of IL with Privileged Teachers: Locomotion

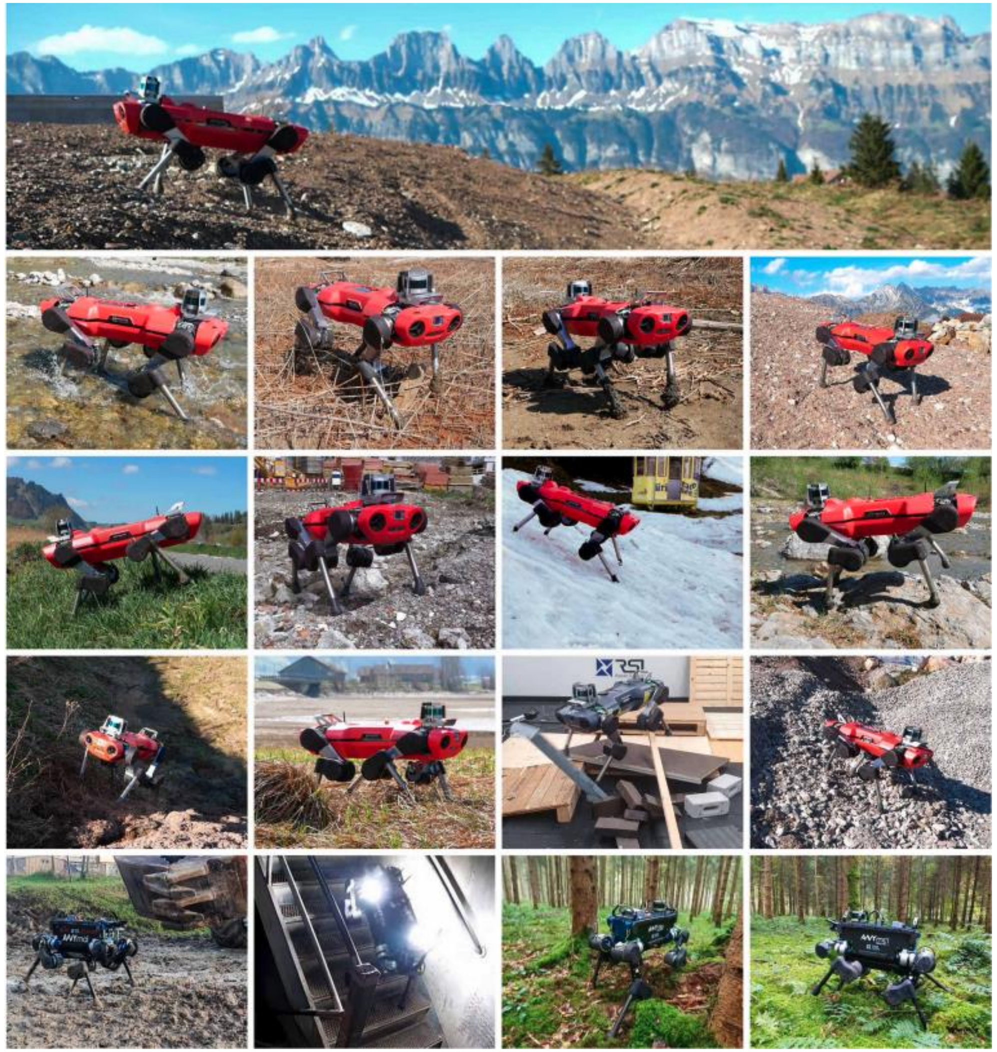
- This privileged learning pipeline is even easier and more useful in simulation!

Learning quadrupedal locomotion over challenging terrain

Joonho Lee[1*], Jemin Hwangbo[1,2], Lorenz Wellhausen[1], Vladlen Koltun[3], Marco Hutter[1]

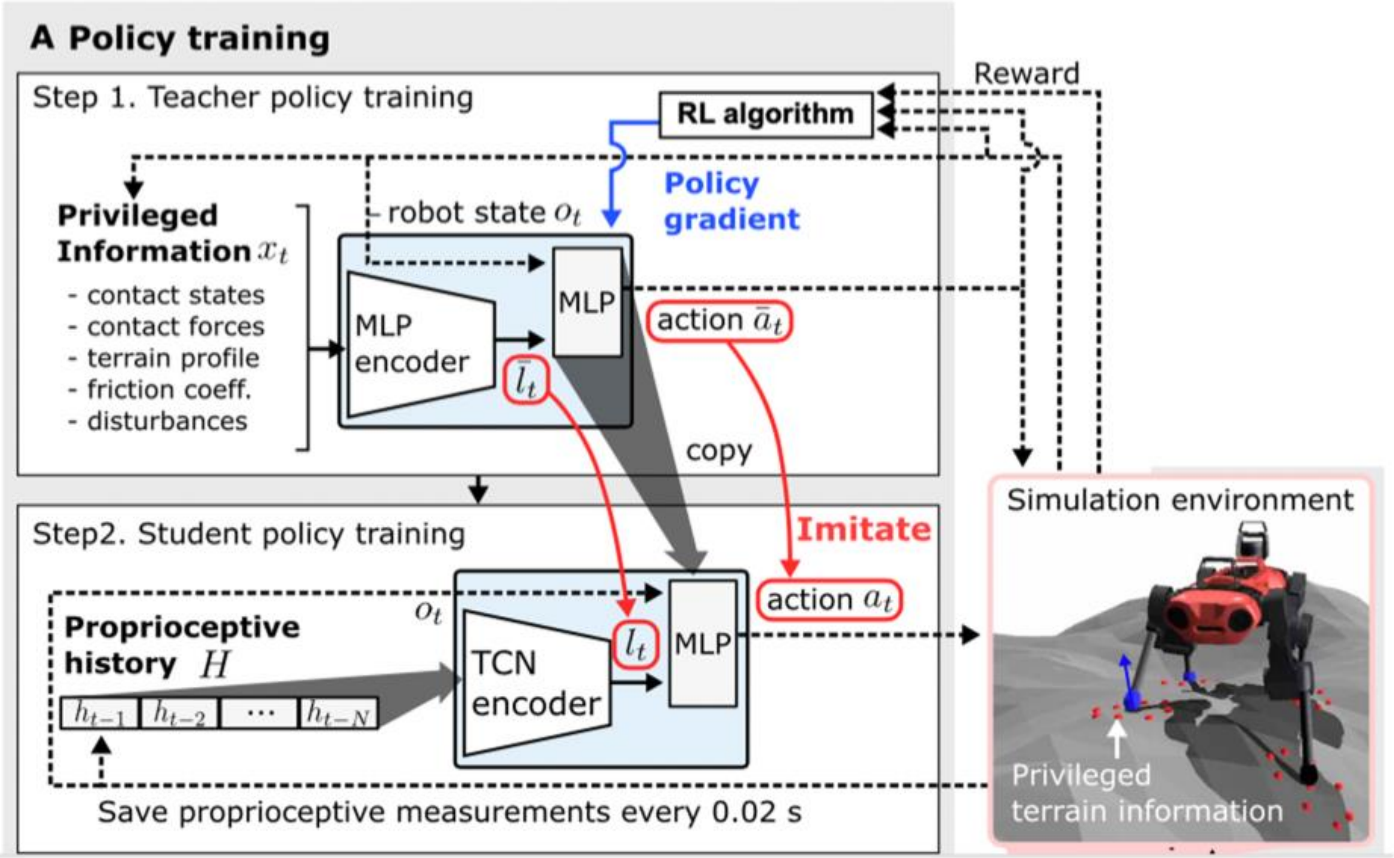
Example of IL with Privileged Teachers: Locomotion

- This privileged learning pipeline is even easier and more useful in simulation!

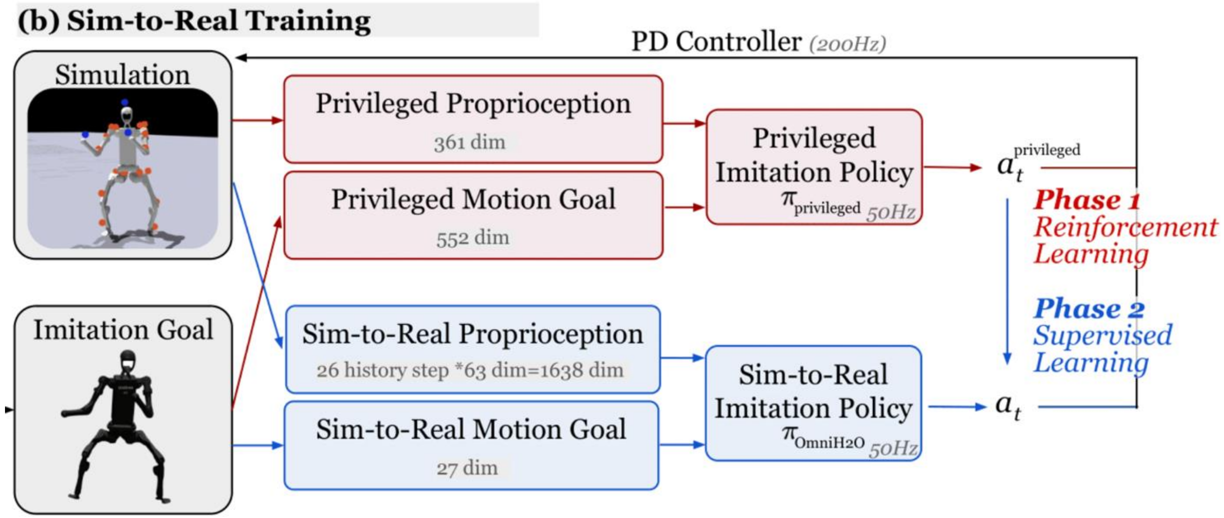


Example of IL with Privileged Teachers: Locomotion

- ❑ Privileged information contains almost everything available in sim (contact, terrain, fraction, disturbance, etc)
- ❑ Students only learn from proprioceptive history (IMU, joint angle, etc)
- ❑ Privileged teacher is trained using RL (PPO, different from the driving example)



Example of IL with Privileged Teachers: Humanoids



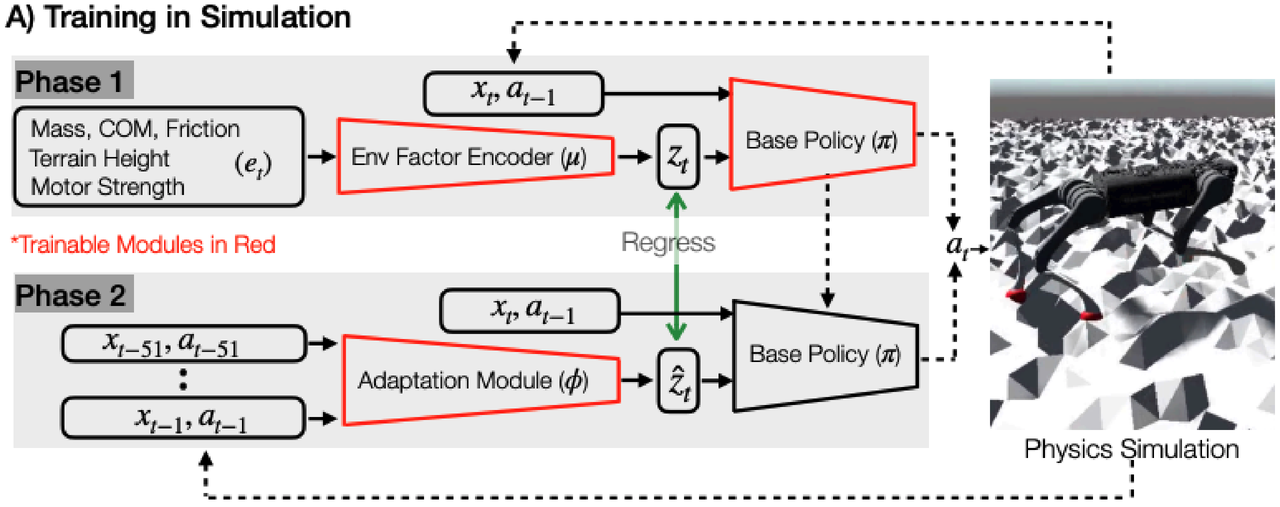
OmniH2O: Universal and Dexterous Human-to-Humanoid Whole-Body Teleoperation and Learning, He & Luo* & He*, et al., CoRL'24*

Some Variants: Student Learning Not in Action Space

- ❑ Student learning doesn't have to happen in the action space
  - RMA: in latent space

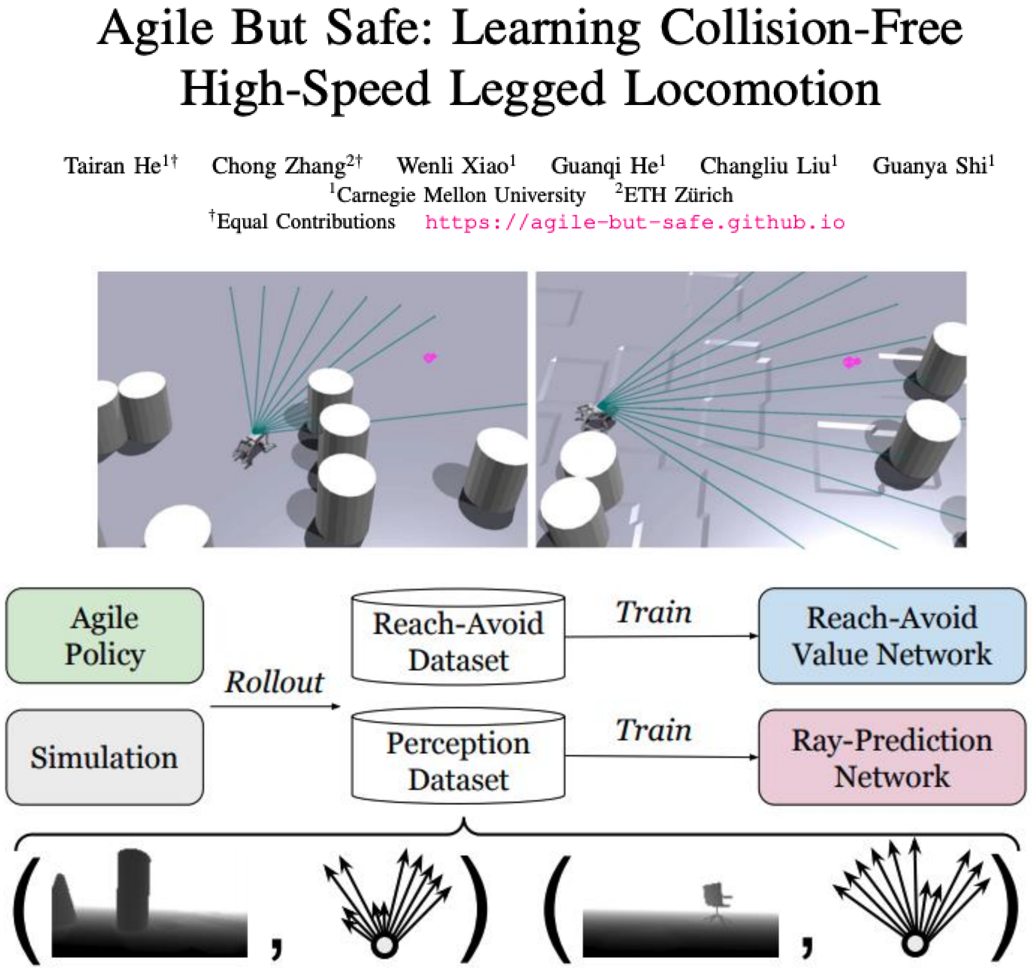
RMA: Rapid Motor Adaptation for Legged Robots

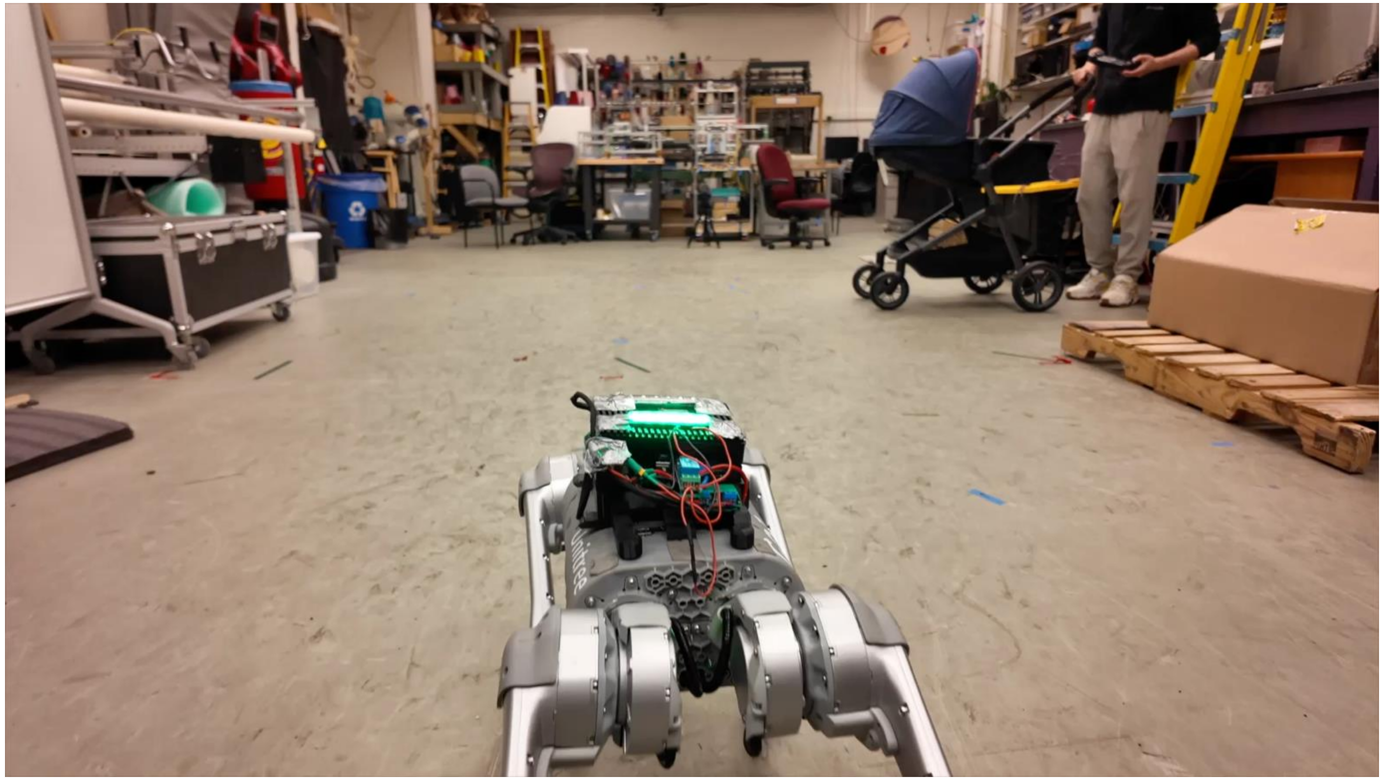
Ashish Kumar, UC Berkeley
Zipeng Fu, Carnegie Mellon University
Deepak Pathak, Carnegie Mellon University
Jitendra Malik, UC Berkeley, Facebook



Some Variants: Student Learning Not in Action Space

- Student learning doesn't have to happen in the action space
  - ABS: student predicts rays





Example of IL with Privileged Teachers: Drone

Deep Drone Acrobatics

Elia Kaufmann*‡, Antonio Loquercio*‡, René Ranftl†, Matthias Müller†, Vladlen Koltun†, Davide Scaramuzza‡

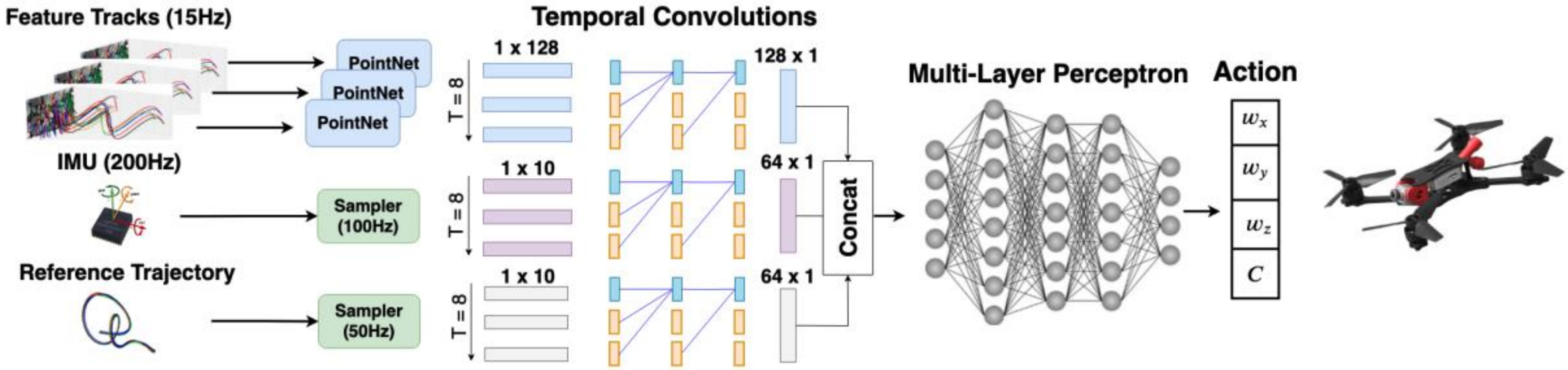
Example of IL with Privileged Teachers: Drone

- ❑ Privileged teacher is an MPC controller
  - Knowing state

$$\pi^* = \min_{\boldsymbol{u}} \Big\{ \boldsymbol{x}[N]^\top \mathcal{Q} \boldsymbol{x}[N] + \sum_{k=1}^{N-1} \left(\boldsymbol{x}[k]^\top \mathcal{Q} \boldsymbol{x}[k] + \boldsymbol{u}[k]^\top \mathcal{R} \boldsymbol{u}[k] \right) \Big\}$$

$$\text{s.t.} \quad \boldsymbol{r}(\boldsymbol{x}, \boldsymbol{u}) = 0$$

$$\boldsymbol{h}(\boldsymbol{x}, \boldsymbol{u}) \leq 0,$$



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- ❑ Imitation learning with privileged teachers
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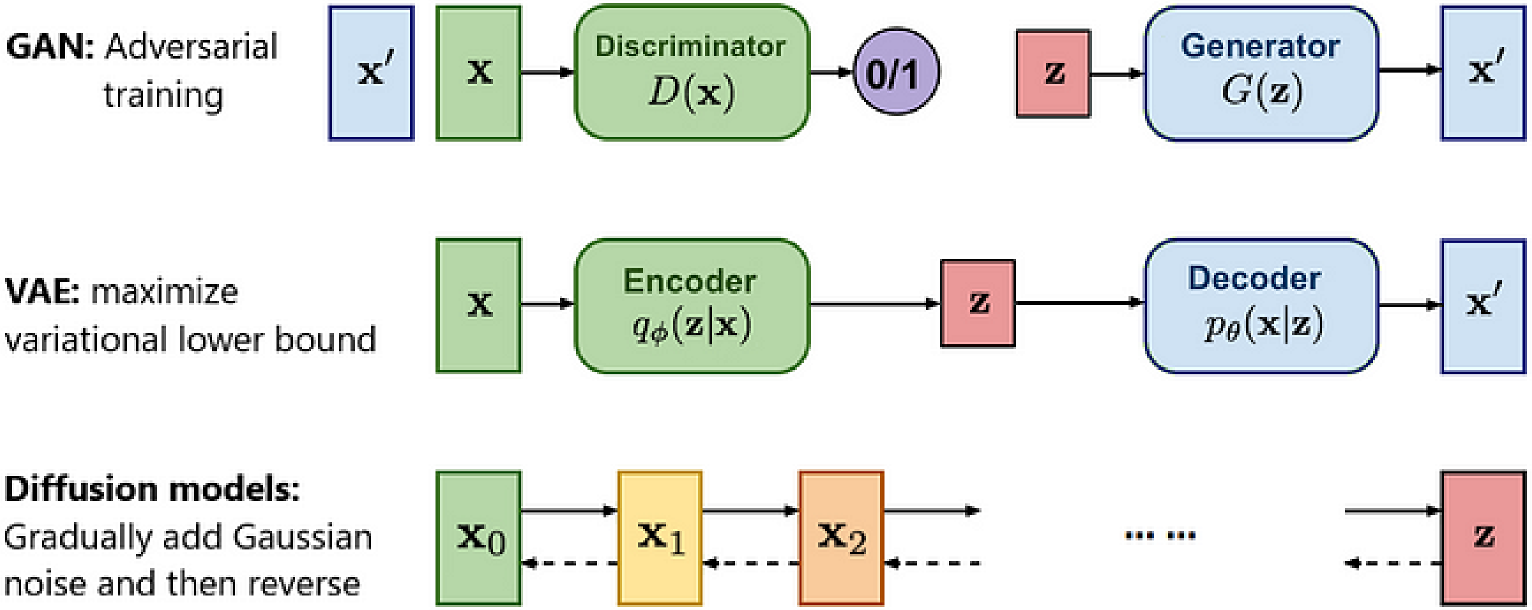
  - Very active research area!

(Some slides are from 22F, 23F, Berkeley CS 285, and CMU 10-703)

Recap: Generative Models

- ❑ What are generative models?
  - Learning: Learn a distribution p_θ that "matches" p_{data}
  - Sampling: Generate novel data, $x_{\text{new}} \sim p_\theta$
- ❑ Almost all types of generative models can be integrated with IL
- ❑ p_{data} is from expert: can be action level, trajectory level, …
- ❑ Often conditional

$\mathbf{x}^{(j)} \sim p_{\text{data}}$
$j = 1, 2, \dots, |\mathcal{D}|$



GAN + Imitation Learning

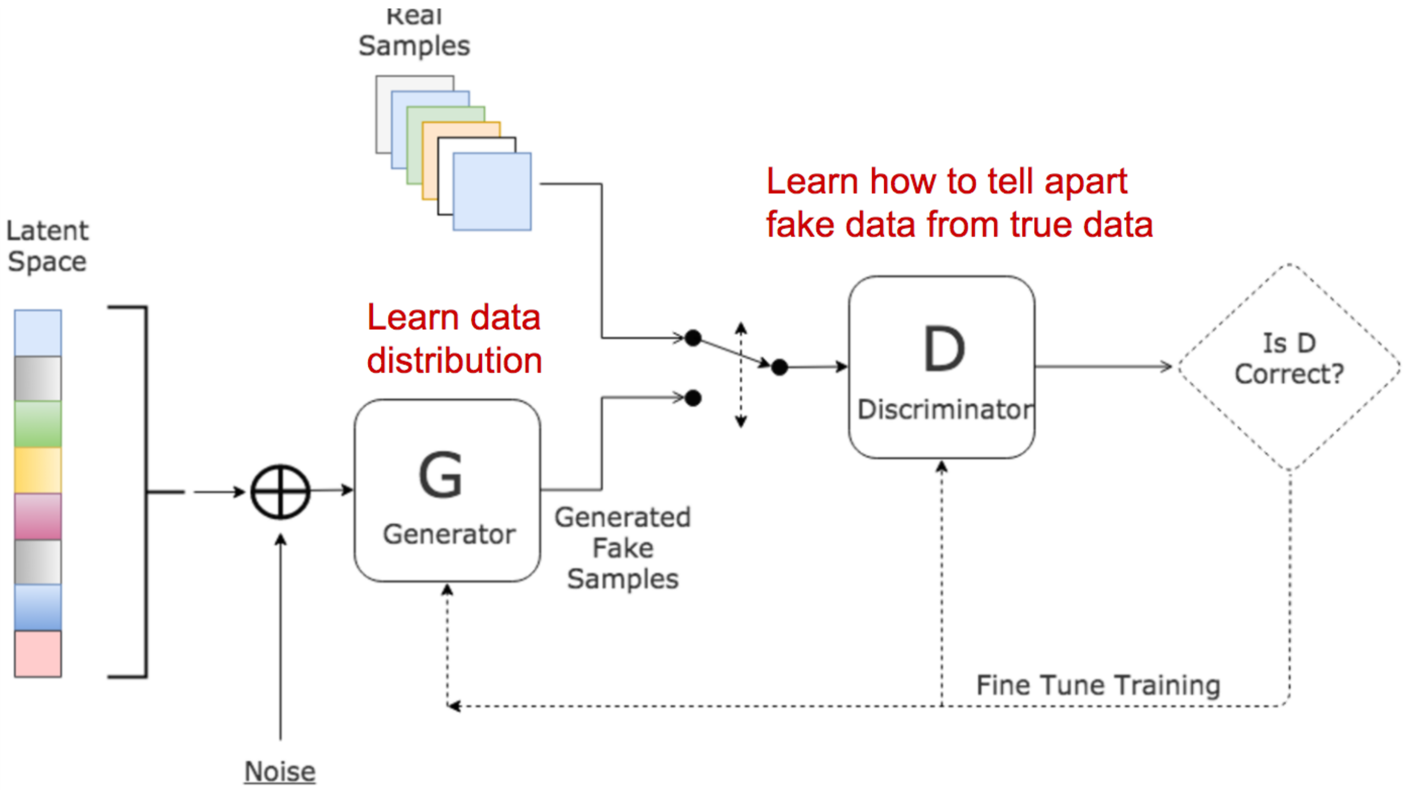
- ❑ Generative Adversarial IL (GAIL)
- ❑ Recap: GAN (lecture 4)
- ❑ GAIL: $x = (s, a)$

Generative Adversarial Imitation Learning

Jonathan Ho
OpenAI
hoj@openai.com

Stefano Ermon
Stanford University
ermon@cs.stanford.edu

$$\min_G \max_D V(D, G) = \mathbb{E}_{\boldsymbol{x} \sim p_{\text{data}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{z} \sim p_{\boldsymbol{z}}(\boldsymbol{z})}[\log(1 - D(G(\boldsymbol{z})))].$$



GAN + Imitation Learning

- GAIL's main idea (repeat the following three steps):
  - Sample trajectory from students
  - Update discriminator, which aims to classify teacher and student
  - Train student policy which aims to minimize the discriminator's accuracy
- Question: can GAIL model multi-modal behavior?

Algorithm 1 Generative adversarial imitation learning

1: **Input:** Expert trajectories $\tau_E \sim \pi_E$, initial policy and discriminator parameters θ_0, w_0
2: **for** $i = 0, 1, 2, \ldots$ **do**
3: Sample trajectories $\tau_i \sim \pi_{\theta_i}$
4: Update the discriminator parameters from w_i to w_{i+1} with the gradient

$$\hat{\mathbb{E}}_{\tau_i}[\nabla_w \log(D_w(s,a))] + \hat{\mathbb{E}}_{\tau_E}[\nabla_w \log(1 - D_w(s,a))] \tag{17}$$

5: Take a policy step from θ_i to θ_{i+1}, using the TRPO rule with cost function $\log(D_{w_{i+1}}(s,a))$. Specifically, take a KL-constrained natural gradient step with

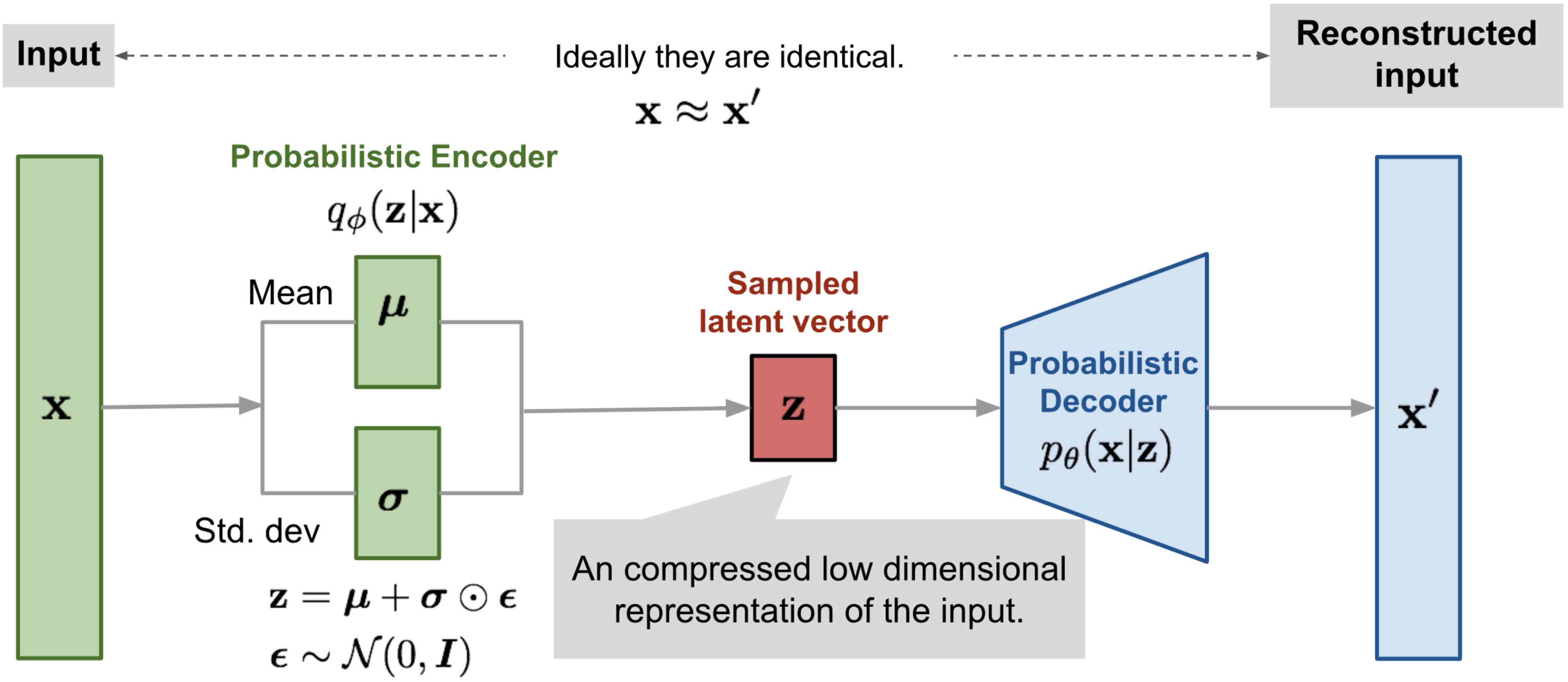
$$\begin{aligned} &\hat{\mathbb{E}}_{\tau_i}\left[\nabla_\theta \log \pi_\theta(a|s) Q(s,a)\right] - \lambda \nabla_\theta H(\pi_\theta), \\ &\text{where } Q(\bar{s},\bar{a}) = \hat{\mathbb{E}}_{\tau_i}[\log(D_{w_{i+1}}(s,a)) \mid s_0 = \bar{s}, a_0 = \bar{a}] \end{aligned} \tag{18}$$

6: **end for**

VAE + Imitation Learning

- Recap: VAE (lecture 4)

$$L_{\mathrm{VAE}}(\theta,\phi) = -\log p_\theta(\mathbf{x}) + D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x})\|p_\theta(\mathbf{z}|\mathbf{x}))$$
$$= -\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\log p_\theta(\mathbf{x}|\mathbf{z}) + D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x})\|p_\theta(\mathbf{z}))$$
$$\theta^*,\phi^* = \arg\min_{\theta,\phi} L_{\mathrm{VAE}}$$

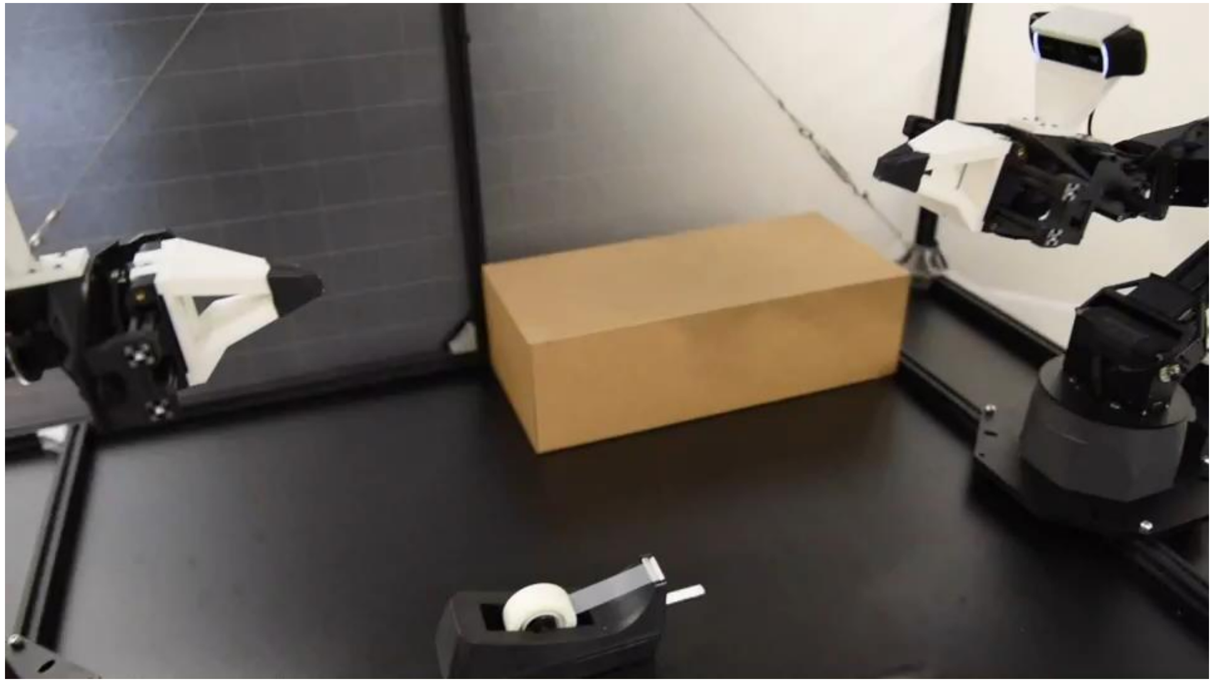


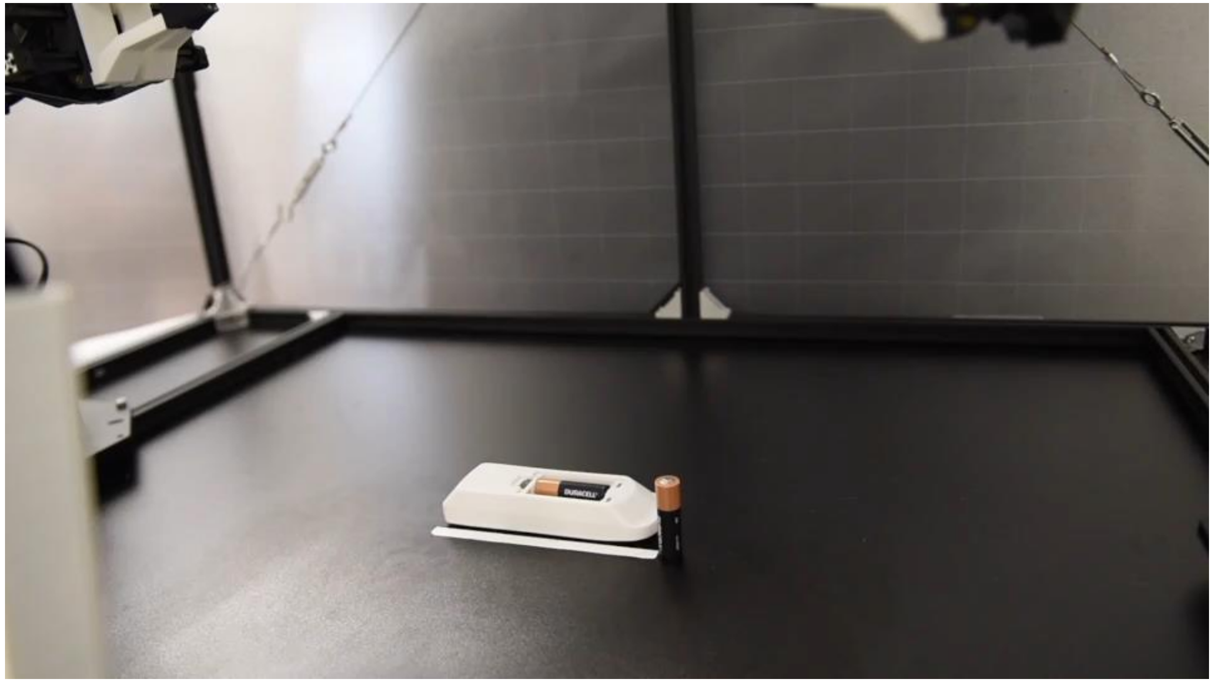
VAE + Imitation Learning

- ❑ Example: Action Chunking with Transformers (ACT)

Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware

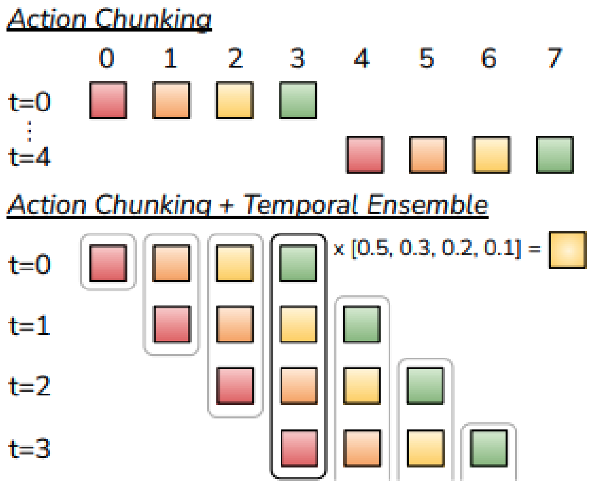
Tony Zhao   Vikash Kumar   Sergey Levine   Chelsea Finn

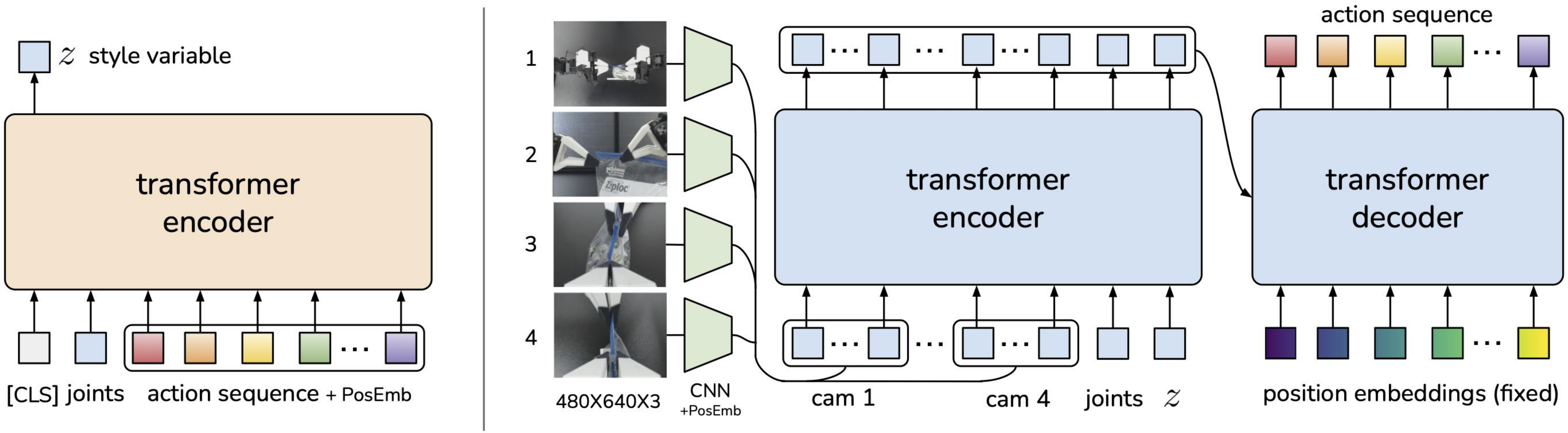




VAE + Imitation Learning

- Example: Action Chunking with Transformers (ACT)
  - Based on CVAE (conditional VAE)
  - Encoder: expert action sequence + observation -> latent
  - Decoder: latent + more observation -> action sequence prediction
  - Key: action chunking + temporal ensemble

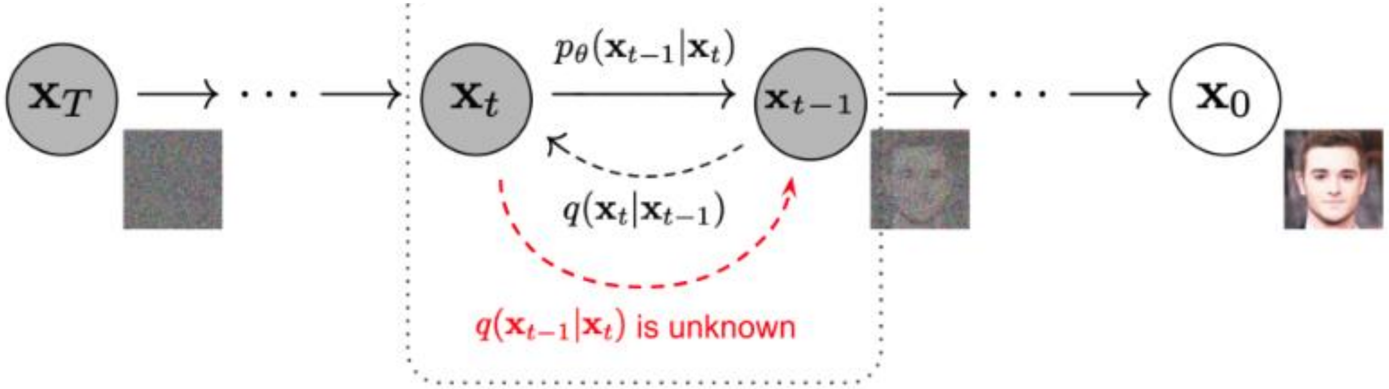




Diffusion Model + Imitation Learning

- Recap: diffusion model (lecture 4)
- **Forward diffusion process**: sample a real data $\mathbf{x}_0$, and gradually corrupt it by iid Gaussians

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \quad q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})$$



- **Reverse diffusion process**: learn Gaussian $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ and then recreate $\mathbf{x}_0$ from $\mathbf{x}_T$

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T)\prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) \quad p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$$

Diffusion Model + Imitation Learning

- Recap: diffusion model (lecture 4)

Algorithm 1 Training

repeat
  $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
  $t \sim \mathrm{Uniform}(\{1, \ldots, T\})$
  $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  Take gradient descent step on
    $\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$
until converged

Algorithm 2 Sampling

$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, \ldots, 1$ **do**
  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
  $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z}$
end for
return $\mathbf{x}_0$

- Connections with **stochastic gradient Langevin dynamics**
  - Concept from physics. It produces samples from $p(\mathbf{x})$ using only its log gradient:

$$\mathbf{x}_t = \mathbf{x}_{t-1} + \frac{\delta}{2}\underbrace{\nabla_{\mathbf{x}} \log p(\mathbf{x}_{t-1})}_{\text{score}} + \sqrt{\delta}\boldsymbol{\epsilon}_t, \quad \text{where } \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

  - When $t \to \infty, \delta \to 0$, the distribution of $\mathbf{x}_t$ equals to the true $p(\mathbf{x})$

Diffusion Model + Imitation Learning

- ❑ Example 1:

IMITATING HUMAN BEHAVIOUR
WITH DIFFUSION MODELS

Tim Pearce, Tabish Rashid, Anssi Kanervisto, Dave Bignell, Mingfei Sun, Raluca Georgescu, Sergio Valcarcel Macua, Shan Zheng Tan, Ida Momennejad, Katja Hofmann, Sam Devlin
Microsoft Research

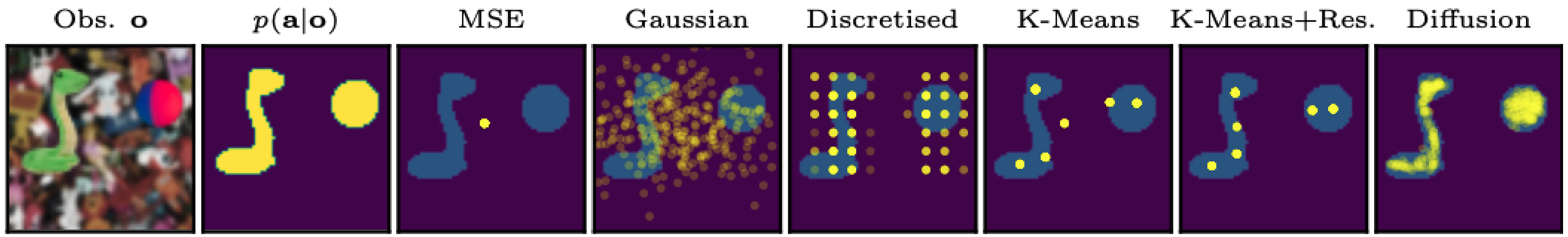
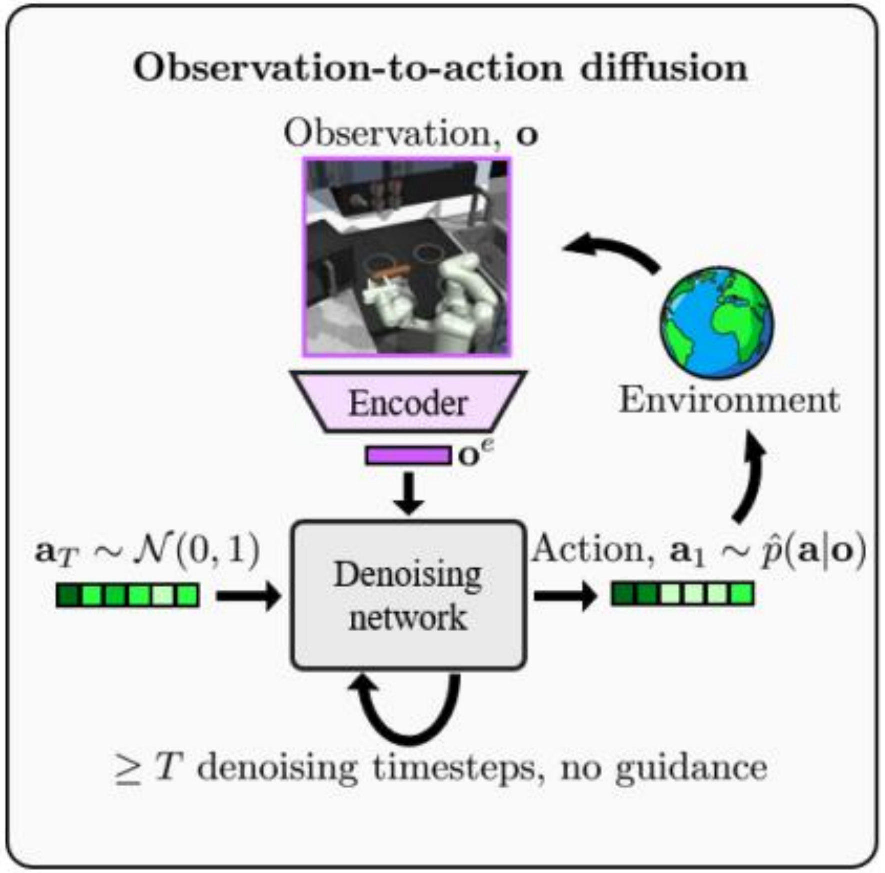


Figure 1: Expressiveness of a variety of models for behaviour cloning in a single-step, arcade claw game with two simultaneous, continuous actions. Existing methods fail to model the full action distribution, $p(\mathbf{a}|\mathbf{o})$, whilst diffusion models excel at covering multimodal & complex distributions.

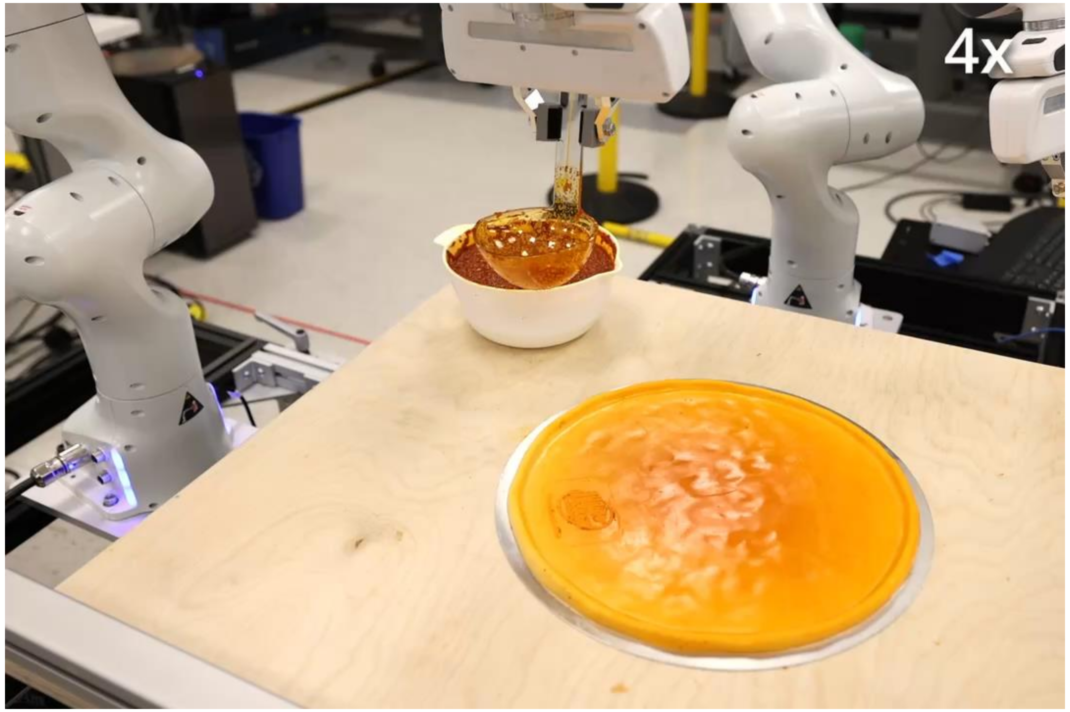


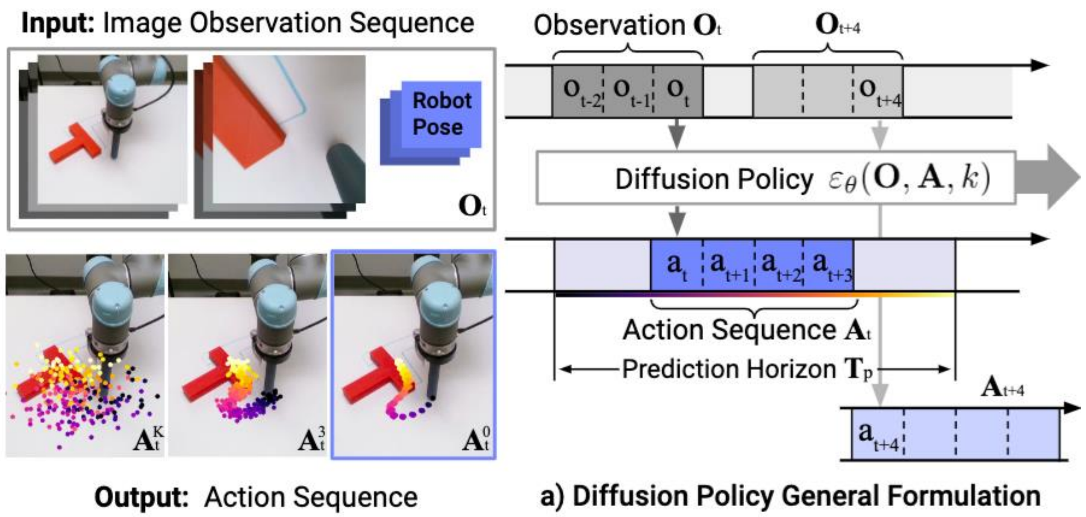
Diffusion Model + Imitation Learning

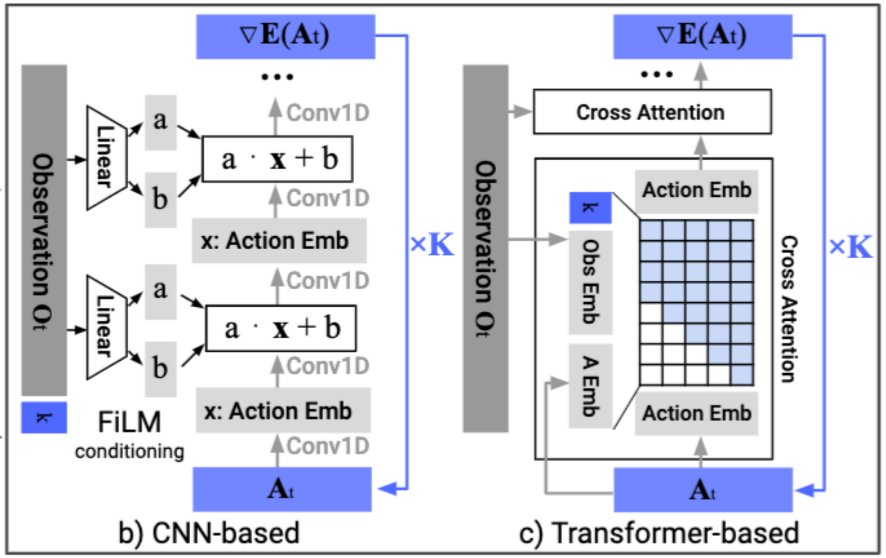
- ❑ Example 2:

Diffusion Policy:
Visuomotor Policy Learning via Action Diffusion

Cheng Chi[1], Siyuan Feng[2], Yilun Du[3], Zhenjia Xu[1], Eric Cousineau[2], Benjamin Burchfiel[2], Shuran Song[1]
[1] Columbia University [2] Toyota Research Institute [3] MIT
https://diffusion-policy.cs.columbia.edu







Diffusion Model + Imitation Learning

- Example 3: diffuser (trajectory-level diffusion)

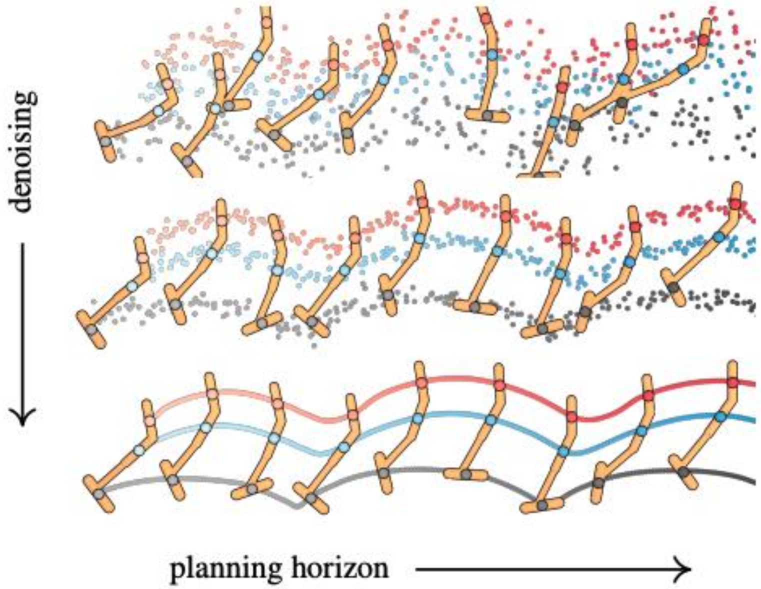
Algorithm 1 Guided Diffusion Planning

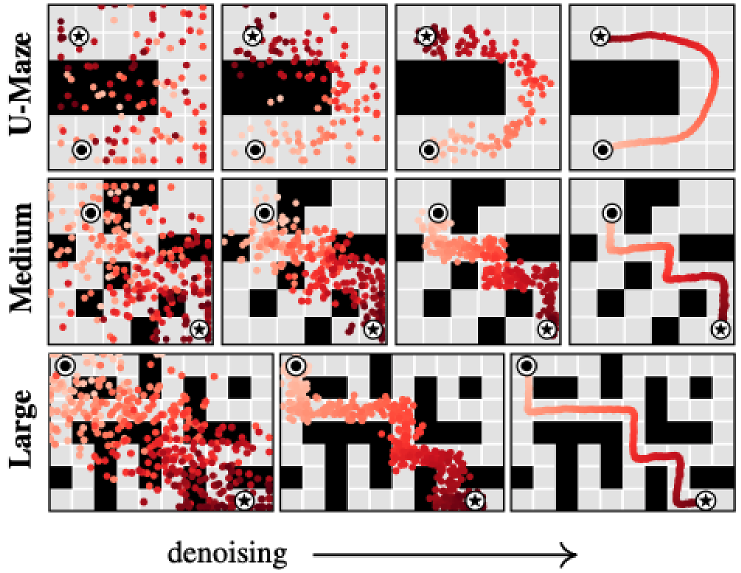
Require Diffuser μ_θ, guide $\mathcal{J}$, scale α, covariances Σ^i
while not done **do**
  Observe state $\boldsymbol{s}$; initialize plan $\boldsymbol{\tau}^N \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$
  for $i = N, \ldots, 1$ **do**
    `// parameters of reverse transition`
    $\mu \leftarrow \mu_\theta(\boldsymbol{\tau}^i)$
    `// guide using gradients of return`
    $\boldsymbol{\tau}^{i-1} \sim \mathcal{N}(\mu + \alpha\Sigma\nabla\mathcal{J}(\mu), \Sigma^i)$
    `// constrain first state of plan`
    $\boldsymbol{\tau}^{i-1}_{\boldsymbol{s}_0} \leftarrow \boldsymbol{s}$
  Execute first action of plan $\boldsymbol{\tau}^0_{\mathbf{a}_0}$

Planning with Diffusion for Flexible Behavior Synthesis

Michael Janner[*1] Yilun Du[*2] Joshua B. Tenenbaum[2] Sergey Levine[1]

$$\boldsymbol{\tau} = \begin{bmatrix} \mathbf{s}_0 & \mathbf{s}_1 & \ldots & \mathbf{s}_T \\ \mathbf{a}_0 & \mathbf{a}_1 & & \mathbf{a}_T \end{bmatrix}$$





Summary: Generative Models + IL

- IL can be integrated with various generative models:
    - GAN
    - VAE
    - Diffusion models
- The learned distribution could be
    - Action level (conditional on observation) $p(a_t|o_t)$
    - State-action (or observation-action) pair $p(s_t, a_t)$
    - A sequence of actions
    - A full state-action trajectory

Thank You!